

# Forcing and its Applications

Prof. Ralf Schindler, Dr. Ben de Bondt

LaTeXed by Changling Liu

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*Any mistakes, inaccuracies, or omissions in these notes are solely my responsibility and should not be attributed to the lecturers. Please feel free to contact me at [cliu4@uni-muenster.de](mailto:cliu4@uni-muenster.de).*

## Contents

|                                                                                                                                                                               |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>0 Basic Knowledge</b>                                                                                                                                                      | <b>3</b>  |
| <b>1 Boolean Value Forcing</b>                                                                                                                                                | <b>6</b>  |
| <b>2 Applications</b>                                                                                                                                                         | <b>15</b> |
| 2.1 A model of ZFC with no "definable" well-order of reals and a model of ZF in which there is a countable set of countable sets of reals whose union is uncountable. . . . . | 15        |
| 2.1.1 A model of ZFC with no "definable" well-order of reals . . . . .                                                                                                        | 15        |
| 2.1.2 A model of ZF in which there is a countable set of countable sets of reals whose union is uncountable . . . . .                                                         | 19        |
| 2.2 Solovay's model: A model of ZF where every set of real is Lebesgue measurable and has the Baire property and perfect set property . . . . .                               | 22        |
| 2.2.1 The BIG Theorem and its LONG Proof . . . . .                                                                                                                            | 29        |

## 0 Basic Knowledge

We begin with a bunch of definitions:

- A **forcing poset** (/forcing/forcing notion) is a triple  $(\mathbb{P}, \leq_{\mathbb{P}}, \mathbb{1}_{\mathbb{P}})(=\mathbb{P})$  such that  $\leq$  is a **pre-order** on the set  $\mathbb{P}$ , elements of  $\mathbb{P}$  are called **conditions**(/forcing conditions). We say that  $q$  is stronger than  $p$  if  $q \leq_{\mathbb{P}} p$ .
- $D \subseteq \mathbb{P}$  is **dense** if  $\forall p \in \mathbb{P}, \exists q \in D(q \leq p)$ .
- $G \subseteq \mathbb{P}$  is a **filter** if  $\forall p \in G, \forall q \in \mathbb{P}(p \leq q \rightarrow q \in G)$  and  $\forall p \in G, \forall q \in G, (\exists r \in G, r \leq p, r \leq q)$ .
- For  $p, q \in \mathbb{P}$ , we say  $p$  and  $q$  are **compatible** if  $\exists r \in \mathbb{P}$  such that  $r \leq p, r \leq q$ , denoted as  $p \parallel q$ . Otherwise it is called **incompatible**, denoted as  $p \perp q$ .
- $\mathbb{P}$  is called **atomless** if  $\forall p \in \mathbb{P}, \exists q, r \leq p(q \neq r \rightarrow q \perp r)$ , a condition  $p \in \mathbb{P}$  is called **atom** if  $\forall q, r \leq p(q \neq r \rightarrow q \parallel r)$ .
- $A \subseteq \mathbb{P}$  is an **antichain** if and only if  $\forall p, q \in A, (p \neq q \rightarrow p \perp q)$ . We say  $\mathbb{P}$  has the **countable chain condition** (c.c.c.) if every antichain is countable.
- Let  $\mathcal{D}$  be a set of dense subsets of  $\mathbb{P}$ , a filter  $G \subseteq \mathbb{P}$  is called  **$\mathcal{D}$ -generic** if  $\forall D \in \mathcal{D}, G \cap D \neq \emptyset$ .

Let  $V$  be a transitive model of ZFC, let  $\mathbb{P} \in V$  be a forcing poset. A filter  $G \subseteq \mathbb{P}$  is  **$V$ -generic** if it is  $\mathcal{D}_V$ -generic where  $\mathcal{D}_V = \{D \subseteq \mathbb{P} : D \text{ dense}\} \cap V$ .

If  $\mathbb{P}$  is atomless then a  $V$ -generic filter can never be an element of  $V$ .

**Example 0.1.** Let  $I, J$  be two non-empty sets, let  $\kappa \leq \bar{I}$  be infinite cardinal. Then  $\text{Fn}_{<\kappa}(I, J)$  is the set of all functions  $p: \text{dom}(p) \rightarrow J$  with  $\text{dom}(p) \subseteq I$  and  $\overline{\text{dom}(p)} < \kappa$ .

Consider the following forcing  $(\mathbb{P}, \leq, \mathbb{1}) : \text{Fn}_{<\kappa}(I, J), \leq = \{(q, p) \in \mathbb{P}^2 : q \supseteq p\}, \mathbb{1} = \emptyset$ .

For each  $i \in I, D_i := \{p \in \mathbb{P} : i \in \text{dom}(p)\}$  is dense. Let  $\mathcal{D} = \{D_i : i \in I\}$ , then  $\mathcal{D}$ -generic filters correspond to functions  $g: I \rightarrow J, G \mapsto \bigcup G, \{p \in \mathbb{P} : p \subseteq g\} \leftrightarrow g$ .

**Lemma 0.2.** Let  $\mathbb{P}$  be a forcing poset and  $\mathcal{D}$  to be a countable set of dense subsets of  $\mathbb{P}$ . Then for every  $p \in \mathbb{P}$ , there exists a  $\mathcal{D}$ -generic filter  $G \subseteq \mathbb{P}$  with  $p \in G$ .

**Definition 0.3** (Forcing Axiom). Let  $\kappa$  be a cardinal and  $\mathbb{P}$  be a forcing poset.  $\text{FA}_{\kappa}(\mathbb{P})$  is the statement that whenever  $\mathcal{D}$  is a set of dense subsets of  $\mathbb{P}$  with  $\overline{\mathcal{D}} \leq \kappa$ , then there exists a  $\mathcal{D}$ -generic filter  $G \subseteq \mathbb{P}$ .

Notice that  $\text{FA}_{\omega}$  holds for every  $\mathbb{P}$ .  $\text{FA}_{\aleph_1}$  fails for certain  $\mathbb{P}$ .

**Example 0.4.** Let  $\mathbb{P} = \text{Fn}_{<\omega}(\omega, \omega_1)$ , consider  $D_i := \{p \in \mathbb{P} : i \in \text{dom}(p)\}$  for  $i < \omega, E_{\alpha} := \{p \in \mathbb{P} : \alpha \in \text{ran}(p)\}$  for  $\alpha < \omega_1$ . It is clear that all  $D_i, E_{\alpha}$ 's are dense. Let  $\mathcal{D} = \{D_i : i < \omega\} \cup \{E_{\alpha} : \alpha < \omega_1\}$ .  $\text{FA}_{\aleph_1}(\mathbb{P})$  implies that there exists a  $\mathcal{D}$ -generic filter  $G \subseteq \mathbb{P}$ , define  $g = \bigcup G: \omega \rightarrow \omega_1$  is surjective, clearly leads to a contradiction.

**Definition 0.5** (Martin's Axiom).  $\text{MA}_{\kappa}$  is the statement that  $\text{FA}_{\kappa}(\mathbb{P})$  holds for all c.c.c. forcings.

Every countable forcing is c.c.c.  $\text{Fn}_{<\omega}(I, J)$  is c.c.c. if both  $I$  and  $J$  are countable.

Note that if  $J$  is uncountable then  $\text{Fn}_{<\kappa}(I, J)$  does not have the c.c.c. since fix any  $i \in I$  then  $\{(i, j) : j \in J\}$  is an uncountable antichain.

**Lemma 0.6.** *Let  $I$  be infinite then  $\text{Fn}_{<\omega}(I, J)$  has the c.c.c. if and only if  $J$  is countable.*

**Lemma 0.7.** *If  $\text{MA}_\kappa$  holds then  $\kappa < 2^{\aleph_0}$ .*

*Proof.* Let  $\mathbb{P} = \text{Fn}_{<\omega}(\omega, 2)$ , this is a c.c.c. forcing. There exists a countable set  $\mathcal{D}_0$  of dense subsets of  $\mathbb{P}$  such that for every  $G \subseteq \mathbb{P}$  a  $\mathcal{D}_0$ -generic filter,  $g = \bigcup G$  is a function  $\omega \rightarrow 2$ . For each  $h : \omega \rightarrow 2$ , define  $D_h := \{p \in \mathbb{P} : p \not\subseteq h\}$ . Notice that  $D_h$  is dense, if  $G \subseteq \mathbb{P}$  is  $\mathcal{D}_0 \cup \{D_h\}$ -generic then  $\bigcup G : \omega \rightarrow 2$  is different from  $h$ . Now suppose  $\text{MA}_{2^{\aleph_0}}$  holds true then there exists a filter  $G \subseteq \mathbb{P}$  which is  $\mathcal{D}_0 \cup \{D_h : h \in 2^\omega\}$ -generic. Then  $g = \bigcup G : \omega \rightarrow 2$  but  $g \neq h$  for all  $h \in 2^\omega$  which clearly leads to a contradiction. ■

**Lemma 0.8.** *Assume  $\text{MA}_\kappa$ . Let  $(A_\alpha : \alpha < \kappa)$  be a sequence of infinite subsets of  $\omega$  such that  $\forall \alpha < \beta < \kappa, A_\alpha \cap A_\beta$  is finite. Then for every  $S \subseteq \kappa$ , there exists  $B \subseteq \omega$  such that  $\forall \alpha < \kappa, B \cap A_\alpha$  is infinite if and only if  $\alpha \in S$ .*

*Proof.* Consider the following forcing  $(\mathbb{P}, \leq, 1)$ :  $p \in \mathbb{P}$  if and only if  $p : \text{dom}(p) \rightarrow 2$  with  $\text{dom}(p) \subseteq \omega$  such that  $\forall \alpha \in S, A_\alpha \cap \text{dom}(p)$  is finite and  $p^{-1''}\{1\}$  is finite,  $p \leq q$  if and only if  $p \supseteq q$ ,  $1_\mathbb{P} = \emptyset$ .

To apply Martin's Axiom, we have to verify this forcing poset has the countable chain condition. Suppose there exists an uncountable antichain  $A \subseteq \mathbb{P}$ , since there are only countably many possibilities for  $p^{-1''}\{1\}$ , by pigeonhole principle, there must be  $p, q \in A$  such that  $p^{-1''}\{1\} = q^{-1''}\{1\}$ . But then consider  $r = p \cup q, r \leq p, r \leq q$  which is clearly a contradiction.

Fix  $S \subseteq \kappa$ , now consider the following sets:

For each  $\beta \in \kappa \setminus S, D_\beta := \{q \in \mathbb{P} : A_\beta \subseteq \text{dom}(q)\}$ . We claim each  $D_\beta$  is dense. Pick any  $p \in \mathbb{P}$ , define  $q$  as follows:

$$q : A_\beta \cup \text{dom}(p) \rightarrow 2, q(i) = \begin{cases} p(i), & \text{if } i \in \text{dom}(p), \\ 0, & \text{otherwise} \end{cases}$$

clearly  $q \leq p, q \in D_\beta$ . By  $A_\alpha \cap A_\beta$  is finite for all  $\alpha < \beta < \kappa$ , this  $q$  is still a condition of the forcing poset.

For each  $\beta \in S, k < \omega, E_{\beta,k} := \{q \in \mathbb{P} : \exists i > k, i \in A_\beta \cap \text{dom}(q) \wedge q(i) = 1\}$ . We claim that each  $E_{\beta,k}$  is dense. Pick  $p \in \mathbb{P}$ , since  $\beta \in S$  so  $A_\beta \cap \text{dom}(p)$  is finite, then  $A_\beta \setminus \text{dom}(p)$  is infinite, so for any  $k < \omega$  we can find some  $i \in A_\beta$  such that  $k < i < \omega$ . Now we define  $q \in \mathbb{P}$ :

$$q : \text{dom}(p) \cup \{i\} \rightarrow 2, q(j) = \begin{cases} p(j), & \text{if } j \in \text{dom}(p), \\ 1, & \text{if } i = j. \end{cases}$$

Clearly  $q \leq p, q \in E_{\beta,k}$ .

Let  $\mathcal{D} = \{D_\beta : \beta \in \kappa \setminus S\} \cup \{E_{\beta,k} : \beta \in S, k < \omega\}$ , by  $\text{MA}_\kappa$ , there exists a  $\mathcal{D}$ -generic filter  $G \subseteq \mathbb{P}$ , denote  $g = \bigcup G : \text{dom}(g) \rightarrow 2$  and set  $B = g^{-1''}\{1\}$ .

Note that for each  $\beta \in S, k < \omega, \exists i < \omega$ , such that  $i > k \wedge i \in A_\beta \cap B$ , this shows " $\Rightarrow$ ". Also for each  $\beta \in \kappa \setminus S$ , there exists  $p \in G$  such that  $A_\beta \subseteq \text{dom}(p)$ . Let  $F := p^{-1}''\{1\}$ , then for any  $i \in A_\beta \setminus F, p(i) = 0$  since  $g$  extends  $p$  so for any  $i \in A_\beta \setminus F, g(i) = 0$  thus  $A_\beta \cap B$  is finite. And this shows " $\Leftarrow$ ". ■

**Corollary 0.9.** *If  $\aleph_0 < \kappa < 2^{\aleph_0}$  then  $\text{MA}_\kappa$  implies  $2^\kappa = 2^{\aleph_0}$ .*

*Proof.* Consider the map  $f: \mathcal{P}(\kappa) \rightarrow \mathcal{P}(\omega)$  by sending  $S$  to  $B$ . To show  $2^\kappa = 2^{\aleph_0}$ , it suffices to verify that  $f$  is an injection. Let  $S_1, S_2$  be two distinct subsets of  $\kappa$ ,  $B_1 = f(S_1), B_2 = f(S_2)$  and set  $S_3 = S_1 \cup S_2$ . By the previous lemma, if  $B_1 = B_2 = B$  then  $\forall \alpha \in S_3, B \cap A_\alpha$  is infinite then leads to a contradiction of the construction of  $S_1, S_2$ . ■

**Definition 0.10.** *MA is the statement that  $\text{MA}_\kappa$  holds for all infinite cardinal  $\kappa < 2^{\aleph_0}$ .*

**Corollary 0.11.** 1.  $\text{MA} \rightarrow 2^\kappa = 2^{\aleph_0}$  for all infinite cardinals  $\kappa < 2^{\aleph_0}$ .

2.  $\text{MA} \rightarrow 2^{\aleph_0}$  is regular (Follows from 1 and König's theorem).

- Does CH imply  $\text{MA}_{\aleph_1}$ ? NO.
- Does CH imply MA? Trivially YES.

**Theorem 0.12** (Solovay, Tennenbaum). *Assume GCH holds. Let  $\kappa$  be an regular cardinal greater than  $\aleph_1$  then there exists a c.c.c. forcing  $\mathbb{P}$  such that  $\mathbb{1}_{\mathbb{P}} \Vdash_{\mathbb{P}} \text{MA} + 2^{\aleph_0} = \kappa$ .*

(But we do not prove this theorem in this lecture.)

Now let  $\mathcal{N}$  be the set of all Lebesgue-null subsets of  $\mathbb{R}^d$ , i.e.  $\mathcal{N} = \{A \subseteq \mathbb{R}^d: \mu(A) = 0\}$ .

**Question:** For what  $\kappa$  the following holds:

$$\text{add}_\kappa: \text{ for every sequence } (N_\alpha: \alpha < \kappa) \text{ with } N_\alpha \in \mathcal{N}, \forall \alpha < \kappa, \bigcup_{\alpha < \kappa} N_\alpha \in \mathcal{N}.$$

**Theorem 0.13.** *For every infinite  $\kappa, \text{MA}_\kappa \rightarrow \text{add}_\kappa$ .*

Observation: Under CH,  $\text{add}_\kappa$  fails for every uncountable  $\kappa$ .

*Proof.* Let  $(N_\alpha: \alpha < \kappa)$  be a sequence of null sets in  $\mathbb{R}^d$  and let  $N = \bigcup_{\alpha < \kappa} N_\alpha$ . To show that  $\mu(N) = 0$ , recall  $\bar{\mu}(N) = \inf\{\mu(\mathcal{O}): \mathcal{O} \subseteq \mathbb{R}^d \text{ open } \wedge N \subseteq \mathcal{O}\}$ .

For any  $\varepsilon > 0$ , it suffices to find some  $\mathcal{O} \subseteq \mathbb{R}^d$  open such that  $N \subseteq \mathcal{O}$  and  $\mu(\mathcal{O}) < \varepsilon$ .

Define the following forcing poset  $(\mathbb{P}, \leq, \mathbb{1})$ :

$$p \in \mathbb{P} \Leftrightarrow p \subseteq \mathbb{R}^d \text{ open } \wedge \mu(p) < \varepsilon; \forall p, q \in \mathbb{P}, q \leq p \Leftrightarrow q \supseteq p; \mathbb{1} = \emptyset.$$

We now claim that  $\mathbb{P}$  has c.c.c., suppose there exists a sequence  $(p_\alpha: \alpha < \omega_1)$  with  $p_\alpha \perp p_\beta, \forall \alpha < \beta < \omega_1$ . By construction of  $\mathbb{P}, p_\alpha \perp p_\beta \Rightarrow \mu(p_\alpha \cup p_\beta) \geq \varepsilon, \forall \alpha < \beta < \omega_1$ .

For  $A, B \subseteq \mathbb{R}^d$ , define  $d(A, B) = \mu(A \Delta B) = \mu(A \cup B) - \mu(A \cap B)$  and  $E \subseteq \omega_1$  uncountable such that  $\forall \alpha \in E, \mu(p_\alpha) \leq \varepsilon - \delta, \delta > 0$  and  $d(p_\alpha, p_\beta) \geq \delta$ , for all  $\alpha < \beta < \omega_1 \wedge \alpha, \beta \in E$ . For each  $\alpha \in E$ , let  $q_\alpha \subseteq p_\alpha$  such that  $q_\alpha$  is a finite union of boxes with rational coordinates and  $\mu(p_\alpha \setminus q_\alpha) \leq \frac{\delta}{10}$ . Then

$\forall \alpha, \beta \in E$  disjoint,  $d(q_\alpha, q_\beta) \geq d(p_\alpha, p_\beta) - d(p_\alpha, q_\alpha) - d(p_\beta, q_\beta) > 0$  thus the map which sends  $\alpha \in E$  to  $q_\alpha$  is an injection as  $d(q_\alpha, q_\beta) > 0$ . But  $E$  is uncountable and there are only countably many  $q_\alpha$ 's, contradiction! Thus  $\mathbb{P}$  has the c.c.c.

Consider  $\mathcal{D} = \{D_\alpha : \alpha < \kappa\}$  where  $D_\alpha = \{q \in \mathbb{P} : N_\alpha \subseteq q\}$ , it is easy to check  $D_\alpha$  is dense. By  $\text{MA}_\kappa$ , let  $G \subseteq \mathbb{P}$  be a  $\mathcal{D}$ -filter, define  $\mathcal{O}_G = \bigcup G$ . Note that  $\mu(\mathcal{O}_G)$  can be written as  $\mu(\bigcup_{n < \omega} p_n) = \sup_{k < \omega} (\mu(\bigcup_{n \leq k} p_n)) \leq \varepsilon$  for some sequence  $(p_n : n < \omega)$  with  $p_n \in G$  for all  $n < \omega$ . And  $\mathcal{O}$  is the open set we desired.  $\blacksquare$

## 1 Boolean-Valued Forcing

Let  $\sigma_{Ba}$  be the first-order signature  $(\wedge, \vee, \neg, 0, 1)$ .

**Definition 1.1.** Let  $S \neq \emptyset$  be a set, an **algebra over  $S$**  is a set  $\mathcal{B} \subseteq \mathcal{P}(S)$  that contains  $S$  and is closed under complements and finite disjunction and conjunction.

**Definition 1.2** (Or Proposition). A **Boolean algebra** is  $\sigma_{Ba}$ -structure  $\mathbb{B} = (B, \wedge, \vee, \neg, 0, 1)$  for which there exists a set  $S \neq \emptyset$  and an algebra  $\mathcal{B}$  over  $S$  such that  $\mathcal{B} \cong (\mathbb{B}, \cap, \cup, S \setminus \cdot, \emptyset, S)$ .

**Definition 1.3.** A Boolean algebra  $\mathbb{B}$  is a  $\sigma$ -structure  $\mathbb{B} = (B, \wedge, \vee, \neg, 0, 1)$  which is a model of the following theory:  $\forall x, y, z \in B$

$$x \vee (y \vee z) = (x \vee y) \vee z \text{ (and its dual)}$$

$$x \vee y = y \vee x \text{ (and its dual)}$$

$$x \vee 0 = x \text{ (and its dual)}$$

$$x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z) \text{ (and its dual)}$$

$$x \wedge \neg x = 0 \text{ (and its dual)}$$

$$0 \neq 1$$

Given a Boolean algebra  $\mathbb{B}$ , define the following relation **orderination**  $\leq_{\mathbb{B}}$  on  $\mathbb{B}$ :

$$\forall a, b \in \mathbb{B}, a \leq_{\mathbb{B}} b \Leftrightarrow a \vee b = b \Leftrightarrow a \wedge b = a \Leftrightarrow a \wedge \neg b = 0 (\Leftrightarrow a \rightarrow b = 1)$$

**Definition 1.4.**  $\mathbb{P} = (\mathbb{P}, \leq)$  is called a **bounded lattice** if

- $\leq$  is a partial order on  $\mathbb{P}$
- $\forall a, b \in \mathbb{P}$  both  $a \wedge b := \inf(\{a, b\})$  and  $a \vee b := \sup(\{a, b\})$  exists.
- $\mathbb{P}$  has a least element  $0_{\mathbb{P}}$  and a greatest element  $1_{\mathbb{P}}$ .

A bounded lattice is called **distributive** if and only if  $\wedge$  and  $\vee$  distribute over each other.

$\mathbb{P}$  is **complemented** if for every  $p \in \mathbb{P}$  there exists  $\neg p \in \mathbb{P}$  such that  $p \wedge \neg p = 0_{\mathbb{P}}, p \vee \neg p = 1_{\mathbb{P}}$ .

Notice that  $\neg p$  may not be unique in a distributive bounded lattice if it exists.

**Lemma 1.5.** • If  $\mathbb{B} = (B, \wedge, \vee, 0, 1)$  is a Boolean algebra then  $(\mathbb{B}, \leq_{\mathbb{B}}, 0, 1)$  is a complemented distributive bounded lattice.

• If  $(\mathbb{P}, \leq)$  is a complemented distributive lattice then  $(P, \wedge, \vee, \neg, 0_{\mathbb{P}}, 1_{\mathbb{P}})$  is a Boolean algebra.

A Boolean algebra  $\mathbb{B}$  is itself a forcing but has an atom 0, we associate to it  $\mathbb{B}^+ = \mathbb{B} \setminus \{0\}$ . "Forcing with  $\mathbb{B}$ " refers to forcing with  $\mathbb{B}^+$ , e.g.  $\mathbb{B}$  has the c.c.c. meant  $\mathbb{B}^+$  has the c.c.c.

**Definition 1.6.** An **ultrafilter** on  $\mathbb{B}$  is a filter  $\mathcal{U} \subseteq \mathbb{B}^+$  that satisfies  $\forall a \in \mathbb{B} ((a \in \mathcal{U}) \vee (\neg a \in \mathcal{U}))$ .

**Definition 1.7.** For a topological space  $X$ , let  $\text{clop}(X)$  be the set of clopen subsets of  $X$ .

Notice that  $\text{clop}(X)$  is an algebra of sets on  $X$ .

**Definition 1.8.** A **stone space** is a topological space which is compact, Hausdorff, zero-dimensional (i.e.  $\text{clop}(X)$  is a base for the topology on  $X$ )

If  $\mathbb{B}$  is a Boolean algebra,  $\text{St}(\mathbb{B}) := \{\mathcal{U} \subseteq \mathbb{B} : \mathcal{U} \text{ is an ultrafilter}\}$  equipped with the topology generated by the sets  $N_a = \{\mathcal{U} \in \text{St}(\mathbb{B}) : a \in \mathcal{U}\}$  for  $a \in \mathbb{B}$ .

Now we check  $\text{St}(\mathbb{B})$  is a stone space.

For Hausdorff: let  $U, V$  be two distinct point in the stone space. There is then some  $a \in \mathbb{B}$  such that  $((a \in U) \wedge (\neg a \in V)) \vee ((a \in V) \wedge (\neg a \in U))$  holds. Without the loss of generality we may assume that  $a \in U$  and  $\neg a \in V$ , then  $U \in N_a$  and  $V \in N_{\neg a}$ . Since we have that  $N_a \cap N_{\neg a} = \emptyset$  thus  $\text{St}(\mathbb{B})$  is Hausdorff.

For compact:

For zero-dimensional:

**Theorem 1.9** (Stone's Representation theorem). (1) For every Boolean algebra  $\mathbb{B}$ ,  $\text{St}(\mathbb{B})$  is a stone space and  $\mathbb{B} \xrightarrow{\sim} \text{clop}(\text{St}(\mathbb{B})), b \mapsto N_b$  and  $a \leq b \Leftrightarrow N_a \subseteq N_b$ .

(2) For every stone space  $X$ ,  $X \xrightarrow{\sim} \text{St}(\text{clop}(X)), x \mapsto \{A \in \text{clop}(X) : x \in A\}$ .

**Proposition 1.10.**  $\text{FA}_{\kappa}(\mathbb{B})$  holds if and only if the intersection of at most  $\kappa$  many open dense subsets of  $\text{St}(\mathbb{B})$  is always non-empty.

*Proof.* "  $\Rightarrow$  " Let  $(\mathcal{O}_{\alpha} : \alpha < \kappa)$  be a sequence of open dense subsets of  $\text{St}(\mathbb{B})$ . Define

$$\text{For all } \alpha < \kappa, D_{\alpha} = \{a \in \mathbb{B} : N_a \subseteq \mathcal{O}_{\alpha}\}$$

Now we claim all the  $D_{\alpha}$ s are dense. Fix some  $\alpha$ , pick any  $b(\neq 0) \in \mathbb{B}$ , since  $\mathcal{O}_{\alpha}$  is open dense so  $\mathcal{O}_{\alpha} \cap N_b \neq \emptyset$ , thus there exist some open  $\mathcal{U}$  such that  $\mathcal{U} \subseteq \mathcal{O}_{\alpha} \cap N_b$ . Then there exists  $a \in \mathbb{B}^+$  such that  $\mathcal{U} \in N_a \subseteq \mathcal{O}_{\alpha} \cap N_b$ , then  $a \in D_{\alpha}$  and  $a \leq_{\mathbb{B}} b$ .

By  $\text{FA}_{\kappa}(\mathbb{B})$ , there exists  $G \subseteq \mathbb{B}^+$ , a  $\{D_{\alpha} : \alpha < \kappa\}$ -generic filter, then  $\{N_a : a \in G\}$  is a family of closed subsets with Finite Intersection Property. Since the space is compact,  $\emptyset \neq \bigcap_{\alpha < \kappa} N_{\alpha}$  and  $\bigcap_{\alpha < \kappa} N_{\alpha} \subseteq \bigcap_{\alpha < \kappa} \mathcal{O}_{\alpha}$ .

"  $\Leftarrow$  " Let  $\mathcal{D}$  be a set of dense subsets of  $\mathbb{B}^+$  with  $|\mathcal{D}| < \kappa$ . For every  $D \in \mathcal{D}$ , define  $\mathcal{O}_D := \bigcup_{b \in D} N_b$ , we claim all  $\mathcal{O}_D$ s are open dense. Openness is clear, now we show the density: Pick any base open set

$N_a \subseteq \text{St}(\mathbb{B})$  for some  $a \in \mathbb{B}$ , since  $D$  is dense in  $\mathbb{B}$ , there exist some  $d \in D$  with  $d \leq a$ , by Stone's Representation theorem, we have  $N_d \subseteq N_a$ . Hence  $\mathcal{O}_D \cap N_a \neq \emptyset$  i.e. is dense in  $\text{St}(\mathbb{B})$ .

By assumption, we have  $\bigcap_{D \in \mathcal{D}} \mathcal{O}_D$  is non-empty, let  $\mathcal{U} \in \bigcap_{D \in \mathcal{D}} \mathcal{O}_D$ . Now we claim  $\mathcal{U} \in \mathbb{B}^+$  is  $\mathcal{D}$ -generic. For any  $D \in \mathcal{D}$  we have  $\mathcal{U} \in \mathcal{O}_D$  then there exists  $b \in D$  such that  $\mathcal{U} \in N_b$ , i.e.,  $b \in \mathcal{U}$ . ■

**Definition 1.11.** A Boolean algebra  $\mathbb{B}$  is **complete** if for all  $S \subseteq \mathbb{B}$ , both  $\sup S = \bigvee S$  and  $\inf S = \bigwedge S$  exist.

**Theorem 1.12** (Forcing Theorem). Let  $M$  be a transitive model of ZFC,  $\mathbb{P} \in M$  be a forcing poset,  $G \subseteq \mathbb{P}$  be  $\mathbb{P}$ -generic over  $M$ ,  $\varphi(\bar{x})$  a formula,  $\tau_1, \dots, \tau_n \in M^{\mathbb{P}}$ , then

$$\exists p \in G (p \Vdash \varphi(\bar{\tau})) \Leftrightarrow M[G] \models \varphi(\tau_1^G, \dots, \tau_n^G).$$

Now we recast the general setting of forcing in the language of complete Boolean algebras and Boolean values.

Let  $\mathbb{B}$  be a complete Boolean algebra. Define sets  $V_\alpha^{\mathbb{B}}$  by recursion on  $\alpha \in \text{Ord}$ :

$$V_\alpha^{\mathbb{B}} = \left\{ \tau : \left( \tau \text{ is a function with } \text{dom}(\tau) \subseteq \bigcup_{\beta < \alpha} V_\beta^{\mathbb{B}} \right) \wedge (\text{ran}(\tau) \subseteq \mathbb{B}) \right\}$$

Define  $V^{\mathbb{B}} = \bigcup_{\alpha \in \text{Ord}} V_\alpha^{\mathbb{B}}$ , the elements of  $V_\alpha^{\mathbb{B}}$  are "the  $\mathbb{B}$ -names of rank  $\leq \alpha$ ".

Define for each formula  $\varphi$  and  $\mathbb{B}$ -names  $\tau_1, \dots, \tau_n$ , we associate an element  $\|\varphi(\tau_1, \dots, \tau_n)\|$  which we call it "Boolean value", this proceeds by recursion on the complexity of formulas:

First for atomic case, we define recursively on  $(\text{rk}(\tau_1), \text{rk}(\tau_2))$  for every pair  $(\tau_1, \tau_2)$  of  $\mathbb{B}$ -names, the Boolean values:  $\|\tau_1 \in \tau_2\|, \|\tau_1 = \tau_2\|, \|\tau_1 \subseteq \tau_2\|$ .

Here "by recursion" we meant  $(\alpha_1, \beta_1) \leq (\alpha_2, \beta_2)$  if and only if  $((\max(\alpha_1, \beta_1) < \max(\alpha_2, \beta_2)) \vee ((\max(\alpha_1, \beta_1) = \max(\alpha_2, \beta_2) \wedge (\alpha_1, \beta_1) \leq_{\text{lex}} (\alpha_2, \beta_2)))$ .

$$\|\tau_1 \in \tau_2\| = \bigvee_{\sigma \in \text{dom}(\tau_2)} (\|\tau_1 = \sigma\| \wedge \tau_2(\sigma)), \|\tau_1 \subseteq \tau_2\| = \bigwedge_{\sigma \in \text{dom}(\tau_1)} (\tau_1(\sigma) \rightarrow \|\sigma \in \text{dom}(\tau_2)\|)$$

and  $\|\tau_1 = \tau_2\| = \|\tau_1 \subseteq \tau_2\| \wedge \|\tau_2 \subseteq \tau_1\|$ .

Then we extend this to all  $\varphi(\bar{\nu})$ :

$$\|\neg \varphi(\bar{\tau})\| = \neg \|\varphi(\bar{\tau})\|, \|\varphi \wedge \psi(\bar{\tau})\| = \|\varphi(\bar{\tau})\| \wedge \|\psi(\bar{\tau})\|, \|\exists x \psi(x, \bar{\tau})\| = \bigvee_{\sigma \in V^{\mathbb{B}}} \|\psi(\sigma, \bar{\tau})\|$$

**Definition 1.13** (Forcing Relation). Let  $b \in \mathbb{B}^+$ ,  $\varphi(\bar{\nu})$  be a formula and  $\tau_1, \dots, \tau_n \in V^{\mathbb{B}}$ .

We define  $b \Vdash^{\mathbb{B}} \varphi(\bar{\tau})$  if and only if  $b \leq \|\varphi(\bar{\tau})\|$ .

Note that  $\|\varphi(\bar{\tau})\| = \bigvee \{b \in \mathbb{B}^+ : b \Vdash^{\mathbb{B}} \varphi(\bar{\tau})\}$ .

We defined before  $V^{\mathbb{P}}$  and  $\Vdash^{\mathbb{P}}$  for every forcing  $\mathbb{P}$ , in case  $\mathbb{P} = \mathbb{B}^+$ , for a complete Boolean algebra  $\mathbb{B}$ , both approach are inter-translatable, and for every  $\tau \in V^{\mathbb{P}}$ , there exists  $\tau' \in V^{\mathbb{B}^+}$  such that  $\mathbb{1}_{\mathbb{P}} \Vdash \tau = \tau'$ .

**Theorem 1.14.** Let  $M$  be a transitive model of ZFC,  $\mathbb{B} \in M$  be a complete Boolean algebra,  $G \subseteq \mathbb{B}^+$  be a generic filter over  $M$ ,  $\varphi(\bar{\nu})$  be a formula and  $\tau_0, \dots, \tau_n \in M^{\mathbb{B}}$ , then  $\|\varphi(\bar{\tau})\| \in G$  if and only if  $M[G] \models \varphi(\tau_1^G, \dots, \tau_n^G)$ .



Before we proof this theorem, let us introduce some useful lemma.

**Lemma 1.15.** *Let  $M$  be a transitive model of ZFC,  $\mathbb{B} \in M$  be a complete Boolean algebra. For a filter  $\mathcal{F} \subseteq \mathbb{B}^+$ , the followings are equivalent:*

1.  $\mathcal{F} \subseteq \mathbb{B}^+$  is generic over  $M$ .
2.  $(\forall S \subseteq \mathbb{B}^+ \text{ with } S \in M)(S \subseteq \mathcal{F} \text{ if and only if } \bigwedge S \in \mathcal{F} \text{ and } \mathcal{F} \text{ is an ultrafilter})$ .
3.  $(\forall S \subseteq \mathbb{B}^+ \text{ with } S \in M)(\mathcal{F} \cap S \neq \emptyset \text{ if and only if } \bigvee S \in \mathcal{F})$ .

*Proof.* See Exercise Sheet 2. ■

**Lemma 1.16.** *For all  $\tau_1, \tau_2 \in M^{\mathbb{B}}$ ,*

$$\|\tau_1 \in \tau_2\| \in G \Leftrightarrow \tau_1^G \in \tau_2^G; \|\tau_1 \subseteq \tau_2\| \in G \Leftrightarrow \tau_1^G \subseteq \tau_2^G; \|\tau_1 = \tau_2\| \in G \Leftrightarrow \tau_1^G = \tau_2^G$$

*Proof.* By recursion on  $(\text{rk}(\tau_1), \text{rk}(\tau_2))$ :

$$\begin{aligned} \|\tau_1 \in \tau_2\| \in G &\Leftrightarrow \bigvee_{\sigma \in \text{dom}(\tau_2)} (\|\sigma = \tau_1\| \wedge \tau_2(\sigma)) \in G \Leftrightarrow (\exists \sigma \in \text{dom}(\tau_2)) ((\sigma^G = \tau_1^G) \wedge (\tau_2(\sigma) \in G)) \Leftrightarrow \tau_1^G \in \tau_2^G \\ \|\tau_1 \subseteq \tau_2\| \in G &\Leftrightarrow \left( \bigwedge_{\sigma \in \text{dom}(\tau_1)} (\tau_1(\sigma) \rightarrow \|\sigma \in \text{dom}(\tau_2)\|) \right) \in G \Leftrightarrow (\forall \sigma \in \text{dom}(\tau_1)) ((\tau_1(\sigma) \rightarrow \|\sigma \in \tau_2\|) \in G) \Leftrightarrow \\ &(\forall \sigma \in \text{dom}(\tau_1)) (\tau_1(\sigma) \in G \rightarrow (\sigma^G \in \tau_2^G)) \Leftrightarrow \tau_1^G \subseteq \tau_2^G \end{aligned}$$
■

*proof of the theorem.* By induction on the complexity of  $\varphi$ .

Atomic case: follows from the previous lemma:

$$\begin{aligned} (\varphi \equiv \neg \psi) \quad \|\varphi(\bar{\tau})\| \in G &\Leftrightarrow \|(\neg \psi)(\bar{\tau})\| \in G \Leftrightarrow (\neg \|\psi(\bar{\tau})\|) \in G \\ &\Leftrightarrow \|\psi(\bar{\tau})\| \notin G \xleftrightarrow{\text{IH}} M[G] \not\models \psi(\bar{\tau}^G) \Leftrightarrow \neg(M[G] \models \psi(\bar{\tau}^G)) \\ &\Leftrightarrow M[G] \models \neg \psi(\bar{\tau}^G) \Leftrightarrow M[G] \models \varphi(\bar{\tau}^G) \\ (\varphi \equiv \psi \wedge \xi) \quad \|\varphi(\bar{\tau})\| \in G &\Leftrightarrow \|(\psi \wedge \xi)(\bar{\tau})\| \in G \Leftrightarrow \|\psi(\bar{\tau}) \wedge \xi(\bar{\tau})\| \in G \\ &\Leftrightarrow (\|\psi(\bar{\tau})\| \wedge \|\xi(\bar{\tau})\|) \in G \Leftrightarrow (\|\psi(\bar{\tau})\| \in G) \wedge (\|\xi(\bar{\tau})\| \in G) \\ &\xleftrightarrow{\text{IH}} (M[G] \models \psi(\bar{\tau}^G)) \wedge (M[G] \models \xi(\bar{\tau}^G)) \Leftrightarrow M[G] \models \varphi(\bar{\tau}^G). \\ (\varphi \equiv \exists x \psi(x, \bar{\nu})) \quad \|\varphi(\bar{\tau})\| \in G &\Leftrightarrow \|\exists x \psi(x, \bar{\tau})\| \in G \Leftrightarrow \left( \bigvee_{\sigma \in M^{\mathbb{B}}} \|\psi(\sigma, \bar{\tau})\| \right) \in G \\ &\Leftrightarrow (\exists \sigma \in M^{\mathbb{B}})(\|\psi(\sigma, \bar{\tau})\| \in G) \xleftrightarrow{\text{IH}} (\exists \sigma \in M^{\mathbb{B}})(M[G] \models \psi(\sigma^G, \bar{\tau}^G)) \\ &\Leftrightarrow M[G] \models \exists x \psi(x, \bar{\tau}^G) \Leftrightarrow M[G] \models \varphi(\bar{\tau}^G) \end{aligned}$$
■

Thus we have shown the Forcing Theorem (in the language of Boolean value), and the proof is much easier than the original version.

**Definition 1.17.** For  $\mathbb{P}, \mathbb{Q}$  two forcing posets, a map  $i: \mathbb{P} \rightarrow \mathbb{Q}$  is called **dense embedding** if

1.  $i(1_{\mathbb{P}}) = 1_{\mathbb{Q}}$
2.  $\forall p_1, p_2 \in \mathbb{P} ((p_1 \leq_{\mathbb{P}} p_2) \rightarrow (i(p_1) \leq_{\mathbb{Q}} i(p_2)))$ ,
3.  $\forall p_1, p_2 \in \mathbb{P} ((p_1 \perp p_2) \rightarrow (i(p_1) \perp i(p_2)))$ ,
4.  $\text{ran}(i)$  is dense in  $\mathbb{Q}$ .

Note that  $i(p_1) \perp i(p_2)$  implies  $p_1 \perp p_2$  (follows from 2).

**Theorem 1.18.** For every forcing poset  $\mathbb{P}$ , there exists  $(\mathbb{B}, i)$  (unique upto isomorphic) such that  $\mathbb{B}$  is a complete Boolean algebra and  $i: \mathbb{P} \rightarrow \mathbb{B}^+$  is a dense embedding.  $\mathbb{B}$  is called the Boolean completion of  $\mathbb{P}$ .

*Proof idea 1.* Define  $\mathbb{B} = \{ \bigvee A : A \subseteq \mathbb{P} \}$ . Inspired by the following proposition.

**Proposition 1.19.**  $\mathbb{P} \stackrel{\text{dense}}{\subseteq} \mathbb{B}^+$  complete Boolean algebra then for  $A, B \subseteq \mathbb{P}, \bigvee A \leq \bigvee B$  if and only if  $\forall a \in A (\forall c \in \mathbb{P} \text{ with } c \leq a) (\exists b \in B) (c \parallel b)$ , we say  $A \leq_{\text{pre}} B \Leftrightarrow \bigvee A \leq \bigvee B$ .

*Proof.*  $\Rightarrow$ :  $\forall a \in A$ , we have  $a \leq \bigvee B$  thus  $\forall c \in \mathbb{P}$  with  $c \leq a, c \leq \bigvee B$ . Then  $c = c \wedge \bigvee B = \bigvee_{b \in B} c \wedge b$ , since  $c \neq 0$  thus  $\exists b \in B$  such that  $c \wedge b \neq 0$  which means  $c \parallel b$ .  $\Leftarrow$ : Suppose there exist some  $a \in A$  such that  $a \not\leq \bigvee B$  then  $a \wedge \neg(\bigvee B) \neq 0$ . Since  $\mathbb{P}$  is dense so there exists  $p \in \mathbb{P}$  with  $p \leq a \wedge \neg(\bigvee B)$  thus  $p \wedge \neg(\bigvee B) \neq 0$  but then contradict to our assumption that  $p \parallel b$  for all  $b \in B$ . ■

Now we define an equivalence relation  $\sim$  on  $\mathcal{P}(\mathbb{P})$ ,  $A \sim B \Leftrightarrow (A \leq_{\text{pre}} B) \wedge (B \leq_{\text{pre}} A)$ . Then define  $\bigvee A := [A]_{\sim} \in \mathcal{P}(\mathbb{P}) / \sim$  and  $\bigvee A \leq \bigvee B$  if and only if  $A \leq_{\text{pre}} B$ .

Then  $\mathbb{B}$  is a Boolean completion of  $\mathbb{P}$ . ■

*Proof idea 2.* Using algebra of regular open sets. Let  $(X, \tau)$  be a topological space. An open set  $A \subseteq X$  is regular if  $\text{int}(\overline{A}) = A$ . Note that  $\text{int}(\overline{\text{int}(\overline{A})}) = \text{int}(\overline{A})$ .

For  $A \subseteq X$ , define  $\text{Reg}(A) = \text{int}(\overline{A})$ ,  $\text{ro}(X, \tau) := \{ A \subseteq X : A = \text{Reg}(A) \}$ ,  $A \wedge B = A \cap B$ ,  $A \vee B = \text{Reg}(A \cup B)$ ,  $\neg A = \text{int}(X \setminus A) = X \setminus \overline{A}$ ,  $\bigvee_{i \in I} A_i = \text{Reg}(\bigcup_{i \in I} A_i)$ ,  $\bigwedge_{i \in I} A_i = \text{Reg}(\bigcap_{i \in I} A_i)$ . The order topology  $\tau_{\leq}$  on  $\mathbb{P}$  is the topology generated by the sets  $p \downarrow$  for  $p \in \mathbb{P}$  with  $p \downarrow = \{ q \in \mathbb{P} : q \leq p \}$ . ■

**Lemma 1.20.**  $i: \mathbb{P} \rightarrow \text{ro}(\mathbb{P}, \tau_{\leq}) = \mathbb{B}, p \mapsto \text{int}(\overline{p \downarrow}) = \text{Reg}(p \downarrow)$  is a dense embedding.

*Proof.* •  $i(1_{\mathbb{P}}) = \mathbb{B} = 1_{\mathbb{B}}$ .

- $q \leq p \Rightarrow q \downarrow \subseteq p \downarrow \Rightarrow \text{Reg}(q \downarrow) \subseteq \text{Reg}(p \downarrow)$ .
- $p \perp q \Leftrightarrow (p \downarrow) \cap (q \downarrow) = \emptyset \stackrel{*}{\Leftrightarrow} \text{Reg}(p \downarrow) \cap \text{Reg}(q \downarrow) = \emptyset \stackrel{**}{\Leftrightarrow} \text{Reg}(p \downarrow) \cap \text{Reg}(q \downarrow) = \emptyset \Leftrightarrow \text{Reg}(p \downarrow) \perp \text{Reg}(q \downarrow)$ .  
 $*$ :  $\Leftarrow$ : as  $q \downarrow \subseteq \text{Reg}(q \downarrow)$ ; as for  $\Rightarrow$ :  $(p \downarrow) \cap (q \downarrow) = \emptyset \Rightarrow \overline{p \downarrow} \cap (q \downarrow) = \emptyset$  and  $**$  follows from the same argument.

- Let  $A(\neq \emptyset) \subseteq \mathbb{P}$  be regular open. There is  $p \in \mathbb{P}$  such that  $p \downarrow \subseteq A$ , thus  $\text{Reg}(p \downarrow) \subseteq \text{Reg}(A) = A$ . Hence  $\text{ran}(i)$  is dense in  $\text{ro}(\mathbb{P}, \tau_{\leq}) = \mathbb{B}^+$ . ■

*Proof of the uniqueness of theorem 1.18.* Suppose there exists  $\mathbb{C}$  and a dense embedding  $j: \mathbb{P} \rightarrow \mathbb{C}$ . We want to show there exists two Boolean algebra isomorphism  $g: \mathbb{B} \rightarrow \mathbb{C}, h: \mathbb{C} \rightarrow \mathbb{B}$  and they are hte inverse of each other.

For all  $b \in \mathbb{B}$ , define  $g(b) = \left( \bigvee_{p \in \mathbb{P}, i(p) \leq b} j(p) \right)^{\mathbb{C}}$  by density of  $i$  and similarly define  $h(c) = \left( \bigvee_{p \in \mathbb{P}, j(p) \leq c} i(p) \right)^{\mathbb{B}}$  for all  $c \in \mathbb{C}$ .

Then  $\forall p \in \mathbb{P}, \forall b \in \mathbb{B}$ , we have  $i(p) \leq b \rightarrow j(p) \leq g(b)$  and  $i(p) \perp b \rightarrow j(p) \perp g(b)$ . Hence also  $j(p) \leq g(b) \rightarrow i(p) \leq b$ , if not, then there exists some  $r \in \mathbb{P}$  with  $i(r) \leq i(p)$  and  $i(r) \perp b$ , by the property of dense embedding we have  $r \parallel p$  and thus  $j(r) \parallel j(p), j(r) \perp g(b)$ , but then contradict to  $j(p) \leq g(b)$ .

Hence  $\forall b \in \mathbb{B}, b = \sup\{i(p) : p \in \mathbb{P} \wedge i(p) \leq b\} = \sup\{i(p) : p \in \mathbb{P} \wedge j(p) \leq g(b)\} = h(g(b))$ . ■

So given  $\mathbb{P}, i: \mathbb{P} \rightarrow \mathbb{B}^+$  here  $\mathbb{B}$  is a complete Boolean algebra,  $p \Vdash^{\mathbb{P}} \varphi(\bar{\tau}) \iff i(p) \Vdash^{\mathbb{B}} \varphi(\bar{\sigma})$  for some  $\sigma \in V^{\mathbb{B}}$  corresponding to  $\bar{\tau}$ .

**Proposition 1.21.**

$$\forall \tau_1, \tau_2, \tau_3 \in V^{\mathbb{B}}, \|\tau_1 = \tau_1\| = \mathbb{1}_{\mathbb{B}}, \|\tau_1 = \tau_2\| = \|\tau_2 = \tau_1\|, \|\tau_1 = \tau_2\| \wedge \|\tau_2 = \tau_3\| \leq \|\tau_1 = \tau_3\|,$$

$$\|\tau_1 = \tau_2\| \wedge \|\tau_2 \in \tau_3\| \leq \|\tau_1 \in \tau_3\|, \|\tau_1 \in \tau_2\| \wedge \|\tau_2 = \tau_3\| \leq \|\tau_1 \in \tau_3\|.$$

*Proof.* We show  $\|\tau = \tau\| = 1$  by induction on  $\text{rk}(\tau)$ .  $\|\tau \subseteq \tau\| = \bigwedge_{\sigma \in \text{dom}(\tau)} (\tau(\sigma) \rightarrow \|\sigma \in \tau\|)$ .

Let  $\sigma \in \text{dom}(\tau)$  then  $\tau(\sigma) \rightarrow \|\sigma \in \tau\| \equiv \neg \tau(\sigma) \vee \|\sigma \in \tau\| = \neg \tau(\sigma) \vee \bigvee_{\pi \in \text{dom}(\tau)} (\tau(\pi) \wedge \|\sigma = \pi\|)$ , by induction hypothesis,  $\bigvee_{\pi \in \text{dom}(\tau)} (\tau(\pi) \wedge \|\sigma = \pi\|) \geq \pi(\sigma)$  thus  $\tau(\sigma) \rightarrow \|\sigma \in \tau\| \geq \neg \tau(\sigma) \vee \tau(\sigma) = 1$ . ■

**Lemma 1.22.** Let  $\mathbb{P}$  be a forcing poset and  $A \subseteq \mathbb{P}$  be an antichain. For each  $q \in A$ , let  $\sigma_q$  be a  $\mathbb{P}$ -name. There then exists a  $\mathbb{P}$ -name  $\tau$  such that  $\forall q \in A, q \Vdash \tau = \sigma_q$ .

*Proof.* Define  $\tau = \bigcup_{q \in A} \bigcup_{(\pi, r) \in \sigma_q} \{(\pi, r') : (r' \in \mathbb{P}) \wedge (r' \leq q, r)\}$ . Let  $q \in A$  and  $G \subseteq \mathbb{P}$  be  $\mathbb{P}$ -generic over  $V$  with  $q \in G$ . We claim  $\tau^G = \sigma_q^G$ .

For  $\subseteq$ : If  $a \in \tau^G$  then  $(\exists \hat{q} \in A)(\exists (\pi, r) \in \sigma_{\hat{q}})(\exists r' \leq \hat{q}, r)$  with  $r' \in G$  and  $a = \pi^G$ . So let  $\hat{q}, (\pi, r)$  and  $r'$  be such. Then  $\hat{q} \in G, r \in G, r \leq q$  thus  $q \in G$ , since  $q, \hat{q}$  both in the antichain so  $q = \hat{q}$ . Hence  $a = \pi^G \in \sigma_q^G$ .

For  $\supseteq$ : If  $a \in \sigma_q^G$  then there exists  $(\pi, r) \in \sigma_q$  with  $r \in G$ , since  $q, r$  both in the generic filter  $G$  so there exists  $r' \in G$  with  $r' \leq r, q$  so  $a \in \tau^G$  by definition. ■

**Theorem 1.23.** Let  $\varphi(x_0, \dots, x_n, x)$  be a formula,  $\tau_0, \dots, \tau_n$  be  $\mathbb{P}$ -names,  $p \in \mathbb{P}$ , then  $p \Vdash \exists x \varphi(\tau_0, \dots, \tau_n, x)$  if and only if there exists  $\sigma \in V^{\mathbb{P}}$  such that  $p \Vdash \varphi(\tau_0, \dots, \tau_n, \sigma)$ .

*Proof.* For  $\Leftarrow$  it is clear. Now let us show  $\Rightarrow$ , let  $\mathcal{G}$  consists of all  $A \subseteq p \downarrow$  such that  $A$  is an antichain and  $\forall q \in A, \exists \sigma \in V^{\mathbb{P}}, q \Vdash \varphi(\bar{\tau}, \sigma)$ . We claim  $\mathcal{G} \neq \emptyset$ : let  $q \leq p$  and  $G \subseteq \mathbb{P}$  be a  $\mathbb{P}$ -generic filter with

$q \in G$ . Then  $V[G] \models \exists x \varphi(\bar{\tau}^G, x)$ . So for some  $\sigma \in V^{\mathbb{P}}$ ,  $V[G] \models \varphi(\bar{\tau}^G, \sigma^G)$ , by forcing theorem there exists some  $r \in G$  such that  $r \Vdash \varphi(\bar{\tau}, \sigma)$ , then we can find some  $r' \leq r, q$  with  $r' \Vdash \varphi(\bar{\tau}, \sigma)$ . **But this only explained why there exists some condition  $q$  below  $p$  and some  $\mathbb{P}$ -name  $\sigma$  such that  $q$  forces  $\varphi(\bar{\tau}, \sigma)$ , but why there exist an antichain which consist of such conditions?**

Apply Zorn's lemma on  $\mathcal{G}$ , we obtained an maximal element  $A$  of  $(\mathcal{G}, \subseteq)$ . For each  $q \in A$ , fix some  $\sigma_q \in V^{\mathbb{P}}$  such that  $q \Vdash \varphi(\bar{\tau}, \sigma_q)$ . Apply the previous lemma, choose some  $\sigma \in V^{\mathbb{P}}$  such that  $q \Vdash \sigma_q = \sigma$  for all  $a \in A$ .

Now we claim  $p \Vdash \varphi(\bar{\tau}, \sigma)$ . If not, then find  $r \in p \downarrow$  such that  $r \Vdash \neg \varphi(\bar{\tau}, \sigma)$ . Then  $r \perp q$  for all  $q \in A$ , find  $r' \leq r, \sigma_{r'} \in V^{\mathbb{P}}$  such that  $r' \Vdash \varphi(\bar{\tau}, \sigma_{r'})$ . Then contradict to the maximality of  $A$ .  $\blacksquare$

**Definition 1.24.**  $D \subseteq \mathbb{P}$  is called **predense** if  $\forall p \in \mathbb{P}, \exists d \in D, d \parallel p$ .

**Definition 1.25.** Let  $\mathbb{P}, \mathbb{Q}$  be forcing posets,  $i: \mathbb{P} \rightarrow \mathbb{Q}$  is a **complete embedding** if

1.  $i(1_{\mathbb{P}}) = 1_{\mathbb{Q}}$ ,
2. (\*) If  $A \subseteq \mathbb{P}$  is an maximal antichain then so is  $i''A \subseteq \mathbb{Q}$ ,
3.  $(\forall p, q \in \mathbb{P})(((p \leq_{\mathbb{P}} q) \rightarrow (i(p) \leq_{\mathbb{Q}} i(q))) \wedge ((p \perp q) \rightarrow (i(p) \perp i(q))))$  holds.

And we can replace 2 by " (\*\*) If  $D \subseteq \mathbb{P}$  is predense then so is  $i''D \subseteq \mathbb{Q}$ ".

Now let us check the equivalence: For  $(*) \rightarrow (**)$ , let  $D \subseteq \mathbb{P}$  be predense, now suppose there exists  $q \in \mathbb{Q}$  such that  $\forall d \in D, i(d) \perp q$ . Set  $D \downarrow := \{p \in \mathbb{P}: (\exists d \in D)(p \leq d)\}$ , since  $D$  is predense so  $D \downarrow$  is open dense. Let  $A \subseteq D \downarrow$  be an maximal antichain, actually  $A$  is also maximal in  $\mathbb{P}$  because suppose there exists  $p \in \mathbb{P}$  with  $p \perp a$  for all  $a \in A$ , but then there exists  $d \in D \downarrow$  with  $d \leq p$  and  $d \perp a$  for all  $a \in A$  which contradict to the maximality of  $A$ . Since  $i$  preserves maximal antichain,  $\forall a \in A, \exists d_a \in D$  such that  $a \leq_{\mathbb{P}} d_a$ . Now we have  $i(a) \leq_{\mathbb{Q}} i(d_a)$  and  $i(d_a) \perp q$ , thus  $i(a) \perp q$  then contradict to the maximality of  $i''A$  which leads to a contradiction.

For  $(**) \rightarrow (*)$ , let  $A \subseteq \mathbb{P}$  be an maximal antichain, we want to show  $i''A \subseteq \mathbb{Q}$  is still maximal. Suppose not, there exists  $q \in \mathbb{Q}$  such that  $q \perp i(a)$  for all  $a \in A$ . Set  $A \downarrow := \{p \in \mathbb{P}: \exists a \in A, p \leq a\}$  and it is dense in  $\mathbb{P}$  hence also predense. Since  $i''(A \downarrow) \subseteq \mathbb{Q}$  is predense there exists  $d \in A \downarrow$  such that  $i(d) \parallel q$ . And by construction of  $A \downarrow$ , there exists  $a_d \in A$  such that  $d \leq_{\mathbb{P}} a_d$ , so  $i(d) \leq i(a_d)$ . But  $q \perp i(a_d)$  thus  $q \perp i(d)$  which leads to a contradiction.

**Lemma 1.26.** Let  $i: \mathbb{P} \rightarrow \mathbb{Q}$  be a complete embedding and  $G \subseteq \mathbb{Q}$  be a  $\mathbb{Q}$ -generic filter over  $V$  then  $i^{-1}G \subseteq \mathbb{P}$  is a  $\mathbb{P}$ -generic filter over  $V$ .

*Proof.* Set  $H := i^{-1}G$ . Let  $A \subseteq \mathbb{P}, D \in V$  be an maximal antichain, then  $i''A$  is also an maximal antichain in  $\mathbb{Q}$ , thus  $G \cap i''A \neq \emptyset$ . Say  $q \in i''A \cap G, q = i(r)$  for some  $r \in A$ , then  $r \in A \cap H$ . Let  $p, p' \in H$  then  $i(p), i(p') \in G$ . Denote  $p^{\perp} := \{q \in \mathbb{P}: q \perp p\}$ . Then  $D = p^{\perp} \cup p'^{\perp} \cup (p \downarrow \cap p' \downarrow)$  is dense. So  $H \cap D \neq \emptyset$ , say  $q \in D \cap H$ . If  $q \in p^{\perp}$  (or  $p'^{\perp}$  similar argument holds) then  $q \perp p$  and  $i(q) \perp i(p)$ , but  $i(q), i(p) \in G$  which contradict to  $G$  is a filter thus  $q \in p \downarrow \cap p' \downarrow \cap H$ .  $\blacksquare$

Note: If  $i: \mathbb{P} \rightarrow \mathbb{Q}$  is a complete(dense) embedding and  $G \subseteq \mathbb{Q}$  is  $\mathbb{Q}$ -generic filter over  $V$ . Define  $H = i^{-1''}G$  then  $V[H] \subseteq V[G]$  ( $V[H] = V[G]$ ). Because  $H$  is defined by using  $i$  and  $G$ , and  $i$  is a function in  $V$  so  $H$  must be in  $V[G]$ .

**Definition 1.27.** Define recursively for any  $i_*: V^{\mathbb{P}} \rightarrow V^{\mathbb{Q}}$  by  $i_*(\tau) = \{(i_*(\sigma), i(p)) : (\sigma, p) \in \tau\}$ . By induction, every  $\tau \in V^{\mathbb{P}}, i_*(\tau) \in V^{\mathbb{Q}}$ .

**Lemma 1.28.** If  $i: \mathbb{P} \rightarrow \mathbb{Q}$  is a complete embedding,  $G \subseteq \mathbb{Q}$  is a  $\mathbb{Q}$ -generic filter over  $V$ . Define  $H = i^{-1''}G$  then  $\forall \tau \in V^{\mathbb{P}}, \tau^H = i_*(\tau)^G$ .

Note that every dense embedding is also complete.

Let us check this: let  $f: \mathbb{P} \rightarrow \mathbb{Q}$  be a dense embedding, we want to show  $f$  is complete, i.e., to show if  $D \subseteq \mathbb{Q}$  is predense then  $f''D$  is also predense in  $\mathbb{Q}$ . Let  $q \in \mathbb{Q}$ , we want to find some  $d \in D$  such that  $f(d) \parallel q$ . Since  $\text{ran}(f)$  is dense in  $\mathbb{Q}$  so there exists  $p \in \mathbb{P}$  such that  $f(p) \leq_{\mathbb{Q}} q$  also  $D$  is predense so there exists  $d \in D$  with  $d \parallel p$ . So  $f(d) \parallel f(p)$  and hence  $f(d) \parallel q$ .

**Lemma 1.29.** Let  $i: \mathbb{P} \rightarrow \mathbb{Q}$  be a dense embedding.

1. If  $H \subseteq \mathbb{P}$  is a  $\mathbb{P}$ -generic filter over  $V$  then  $i''H \uparrow$  is  $\mathbb{Q}$ -generic over  $V$ . For  $G := i''H$ , we have  $i^{-1''}G = H$  and  $V[H] = V[G]$ .
2. If  $G \subseteq \mathbb{Q}$  is a  $\mathbb{Q}$ -generic filter over  $V$  then  $i^{-1''}G$  is a  $\mathbb{P}$ -generic over  $V$ . For  $H := i^{-1''}G$ , we have  $i''H \uparrow = G$  and  $V[H] = V[G]$ .
3. For  $p \in \mathbb{P}, \varphi$  a formula,  $\tau_0, \dots, \tau_{n-1} \in V^{\mathbb{P}}, p \Vdash^{\mathbb{P}} \varphi(\tau_0, \dots, \tau_{n-1}) \iff i(p) \Vdash^{\mathbb{Q}} \varphi(i_*(\tau_0), \dots, i_*(\tau_{n-1}))$ .

*Proof.* 1. Let  $G = i''H \uparrow$ , one can directly check that  $G$  is  $\mathbb{Q}$ -generic filter (See Sheet 1 Exercise 2).

Now we show that  $H = i^{-1''}G$ . Since for any  $p \in \mathbb{P} \setminus H$ , there exists  $h \in H, h \perp p$ . This is because otherwise consider the dense set  $D = \{q \in \mathbb{P} : q \leq p \vee q \perp p\}, D \cap H \neq \emptyset$ , say  $h' \in D \cap H$  then  $h' \leq p$  which leads to a contradiction. If  $H' \subseteq \mathbb{P}$  is a  $\mathbb{P}$ -generic filter over  $V$  and  $H' \neq H$  then there exists  $p \in H$  and  $p' \in H'$  such that  $p \perp p'$ . So if  $i^{-1''}G \neq H$  then there exists  $p' \in i^{-1''}G, p \in H$  such that  $p' \perp p$  but then  $i(p') \perp i(p)$  and this contradicts the fact  $i(p') \parallel i(p)$ .

2. Similar.

3. We prove  $\Rightarrow$ . Assume  $p \Vdash^{\mathbb{P}} \varphi(\bar{\tau})$ , we want to show  $i(p) \Vdash^{\mathbb{Q}} \varphi(i_*(\bar{\tau}))$ . Let  $G \subseteq \mathbb{Q}$  be a  $\mathbb{Q}$ -generic filter over  $V$  with  $i(p) \in G$ . Define  $H := i^{-1''}G$  and then  $H$  is  $\mathbb{P}$ -generic over  $V$  with  $p \in H$ , so  $V[H] \models \varphi(\tau_0^H, \dots, \tau_{n-1}^H)$ . Hence  $V[G] \models \varphi(i_*(\tau_0)^G, \dots, i_*(\tau_{n-1})^G)$ .

$\Leftarrow$  is similar. ■

**Remark 1.30.** It follows that if  $\mathbb{P}, \mathbb{Q}$  are forcing posets and there exists a third forcing poset  $\mathbb{R}$  such that  $\mathbb{P}, \mathbb{Q}$  both embed densely into  $\mathbb{R}$ , then forcing with  $\mathbb{P}$  is equivalent to forcing with  $\mathbb{Q}$ .

If  $i: \mathbb{P} \rightarrow \mathbb{R}$  is a complete embedding and  $H$  is  $\mathbb{P}$ -generic over  $V$ ,  $K$  is  $\mathbb{R}$ -generic over  $V$  then  $V \subseteq V[H] \subseteq V[K]$ . In fact  $V[K]$  is a forcing extension of  $V[H]$ , i.e., there exists  $\mathbb{Q} \in V[H]$  and  $G \subseteq \mathbb{Q}$  a  $\mathbb{Q}$ -generic filter over  $V$  such that  $V[K] = V[H][G]$ . Then there exists  $\dot{\mathbb{Q}} \in V^{\mathbb{P}}$  such that  $\mathbb{Q} = \dot{\mathbb{Q}}^H$ .

Suppose  $\mathbb{P}$  is a forcing poset and  $\dot{\mathbb{Q}}$  is a  $\mathbb{P}$ -name such that  $\mathbb{1}_{\mathbb{P}} \Vdash^{\mathbb{P}} \dot{\mathbb{Q}}$  is a forcing poset then  $V \subseteq V[G] \subseteq V[G][H]$ . In fact  $V[G][H]$  is a forcing extension of  $V$ , which means there is a forcing poset  $\mathbb{R} \in V$  and  $K$  a  $\mathbb{R}$ -generic filter over  $V$  such that  $V[K] = V[G][H]$ .

**Definition 1.31.** Let  $\mathbb{P}$  be a forcing poset,  $\dot{\mathbb{Q}}$  a  $\mathbb{P}$ -name for a forcing poset. Then  $\mathbb{P} * \dot{\mathbb{Q}}$  is the following forcing poset:  $(\mathbb{R}, \leq, \mathbb{1}_{\mathbb{R}})$ , where

$$\mathbb{R} = \{(p, \dot{q}) : p \in \mathbb{P}, \dot{q} \in \text{dom}(\dot{\mathbb{Q}}), p \Vdash \dot{q} \in \dot{\mathbb{Q}}\}, \mathbb{1}_{\mathbb{R}} = (\mathbb{1}_{\mathbb{P}}, \mathbb{1}_{\dot{\mathbb{Q}}})$$

and  $(p_1, \dot{q}_1) \leq (p_2, \dot{q}_2)$  if and only if  $(p_1 \leq_{\mathbb{P}} p_2) \wedge (\dot{q}_1 \leq_{\dot{\mathbb{Q}}} \dot{q}_2)$ , here actually the rightmost  $\leq$  is not an order yet, but we can ask what forces this, when we say  $\dot{q}_1 \leq_{\dot{\mathbb{Q}}} \dot{q}_2$  what we meant was  $p_1 \Vdash^{\mathbb{P}} \dot{q}_1 \leq_{\dot{\mathbb{Q}}} \dot{q}_2$ .

We will confuse  $\mathbb{P}$ -name  $\dot{\mathbb{Q}}$  such that  $\mathbb{1}_{\mathbb{P}} \Vdash^{\mathbb{P}} \dot{\mathbb{Q}}$  is a forcing poset with the triple  $(\dot{\mathbb{Q}}, \leq_{\dot{\mathbb{Q}}}, \mathbb{1}_{\dot{\mathbb{Q}}})$  such that

- $\mathbb{1}_{\dot{\mathbb{Q}}} \in \text{dom}(\dot{\mathbb{Q}})$
- $\mathbb{1}_{\mathbb{P}} \Vdash^{\mathbb{P}} \leq_{\dot{\mathbb{Q}}}$  is a preorder on  $\dot{\mathbb{Q}}$
- $\mathbb{1}_{\dot{\mathbb{Q}}} \in \dot{\mathbb{Q}}$  is a largest element of  $(\dot{\mathbb{Q}}, \leq_{\dot{\mathbb{Q}}})$

**Lemma 1.32.** The map  $i: \mathbb{P} \rightarrow \mathbb{P} * \dot{\mathbb{Q}}$  is a complete embedding.

*Proof.* •  $i(\mathbb{1}_{\mathbb{P}}) = (\mathbb{1}_{\mathbb{P}}, \mathbb{1}_{\dot{\mathbb{Q}}})$ .

- $\forall p_1, p_2 \in \mathbb{P}, p_2 \leq p_1 \Leftrightarrow i(p_2) \leq i(p_1)$  and  $p_2 \perp p_1$  implies  $i(p_2) \perp i(p_1) = (p_2, \mathbb{1}_{\dot{\mathbb{Q}}}) \perp (p_1, \mathbb{1}_{\dot{\mathbb{Q}}})$ .
- Let  $D \subseteq \mathbb{P}$  be predense and  $(p, \dot{q}) \in \mathbb{P} * \dot{\mathbb{Q}}$ . Let  $p' \in D$  be such that  $p' \parallel p$  then  $i(p') \parallel (p, \dot{q})$  as there exists  $p'' \leq p, p'$  and  $p'' \Vdash \dot{q} \leq \mathbb{1}_{\dot{\mathbb{Q}}}, p'' \Vdash \dot{q} \leq \dot{q}$  thus  $(p'', \dot{q}) \leq i(p'), (p, \dot{q})$ .

■

**Theorem 1.33.** Let  $\mathbb{P}$  be a forcing poset and  $\dot{\mathbb{Q}}$  be a  $\mathbb{P}$ -name for a forcing poset, then

1. Let  $K$  be a  $\mathbb{P} * \dot{\mathbb{Q}}$ -generic filter over  $V$  and  $G := i^{-1}K, H = \{\dot{q}^G : (p, \dot{q}) \in K\}$  then  $G$  is  $\mathbb{P}$ -generic filter over  $V$  and  $H$  is  $\dot{\mathbb{Q}}^G$ -generic filter over  $V[G]$  and  $V[K] = V[G][H]$ .
2. Let  $G$  be a  $\mathbb{P}$ -generic filter over  $V$  and  $H$  be a  $\dot{\mathbb{Q}}^G$ -generic filter over  $V[G]$ , then  $K = G * H = \{(p, \dot{q}) : p \in G \wedge \dot{q}^G \in H\}$  is  $\mathbb{P} * \dot{\mathbb{Q}}$ -generic over  $V$  and  $V[G][H] = V[K]$ .

## 2 Applications

### 2.1 A model of ZFC with no "definable" well-order of reals and a model of ZF in which there is a countable set of countable sets of reals whose union is uncountable.

#### 2.1.1 A model of ZFC with no "definable" well-order of reals

**Definition 2.1.** A set  $a$  is called **ordinal definable (OD)** if and only if there is a formula  $\varphi$  and ordinals  $\alpha_0, \dots, \alpha_{n-1}$  such that

$$x \in a \iff \varphi(x, \alpha_0, \dots, \alpha_{n-1}).$$

**Theorem 2.2** (Levy-Montague Reflection Principle). For every formula  $\varphi$  there is a club class of  $\alpha$  such that for every such  $\alpha$  and for all  $x_0, \dots, x_{n-1} \in V_\alpha$ ,

$$\varphi(x_0, \dots, x_{n-1}) \iff V_\alpha \models \varphi(x_0, \dots, x_{n-1}).$$

**Lemma 2.3.** A set  $a$  is OD if and only if there is a formula  $\varphi$  and ordinals  $\alpha_0, \dots, \alpha_{n-1} < \alpha$  such that  $a \in V_\alpha$  and for all  $x$ ,

$$x \in a \iff V_\alpha \models \varphi(x, \alpha_0, \dots, \alpha_{n-1}).$$

Hence "Ordinal Definable" is definable.

**Definition 2.4.** A set  $a$  is called **hereditarily ordinal definable (HOD)**,  $a \in \text{HOD}$  if and only if  $\text{TC}(\{a\}) \subseteq \text{OD}$ .

The class of sets which are hereditarily ordinal definable is called HOD, is definable. (**Check**)

**Theorem 2.5** (Gödel).  $\text{HOD} \models \text{ZFC}$ .

*Proof.* Note that HOD is transitive, we then construct HOD as the  $\mathcal{L}_\in$ -model  $(\text{HOD}, \in \upharpoonright_{\text{HOD}})$ .

Then (Ext), (Fund) holds trivially, since  $\omega \in \text{HOD}$ , (Inf) holds, (Pair) and (Union) are easy. For (Pow): let  $a \in \text{HOD}$ , we want to find  $A \in \text{HOD}$  such that  $\text{HOD} \models "A = \mathcal{P}(a)"$ , here  $A$  may not be the set of all subset of  $a$ , instead, we want  $A$  to be the set of all ordinal definable subsets of  $A$ . Let  $A = \{x \subseteq a : x \text{ is OD}\}$ . As  $a \in \text{HOD}$  and "OD" is definable,  $A$  is also OD, hence  $A \in \text{HOD}$ .

For (Aus), let  $a \in \text{HOD}$ ,  $\varphi$  a formula and  $b_0, \dots, b_{n-1} \in \text{HOD}$ . Then  $\bar{a} = \{x \in a : \text{HOD} \models \varphi(x, b_0, \dots, b_{n-1})\}$  is OD, hence  $\bar{a} \in \text{HOD}$ .

For (Rep), we will show axiom schema of collection which is equivalent to (Rep). Pick an ordinal  $\alpha \in \text{HOD}$  big enough such that every  $x \in a$  where  $a \in \text{HOD}$  replaced by some  $\varphi$  by  $y$ ,  $y$  is contained in  $V_\alpha \cap \text{HOD}$ . And clearly  $V_\alpha \cap \text{HOD}$  is in HOD.

For (AC), let  $a$  be ordinal definable, say  $x \in a$  if and only if  $V_\alpha \models \varphi(x, \alpha_0, \dots, \alpha_{n-1})$  (take the least Gödel number of  $\varphi$  which does the job). Since there is a canonical well-order of finite sets of ordinals: let  $b, d$  be finite sets of ordinals, let  $b <^* d$  if and only if  $\max(b \triangle d) \in d$ .  $<^*$  is a well-order (Proof see Sheet 5 Exercise 1). In this fashion, each  $a \in \text{HOD}$  has a canonical "name" where we minimize  $\varphi, \alpha, \{\alpha_0, \dots, \alpha_{n-1}\}$ . If  $a, b$  are both ordinal definable, say  $a \tilde{<} b$  if and only if the canonical "name" of  $a <$  the canonical "name" of  $b$ . Since  $\tilde{<}$  is definable, hence  $\tilde{<}|_a \in \text{HOD}$  for all  $a \in \text{HOD}$ . ■

**Definition 2.6.** Let  $c$  be a set. We say that a set  $a$  is ordinal definable from members of  $c$  if and only if there is a formula  $\varphi$ , ordinals  $\alpha_0, \dots, \alpha_{n-1}$  and elements  $p_0, \dots, p_{m-1} \in c$  such that for all  $x$ ,

$$x \in a \iff \varphi(x, \alpha_0, \dots, \alpha_{n-1}, p_0, \dots, p_{m-1}).$$

**Definition 2.7.** Let  $c$  be a set, we say  $a$  is hereditarily ordinal definable from elements of  $c$  if and only if every element of  $\text{TC}(\{a\})$  is ordinal definable from elements of  $c$ , denoted by  $a \in \text{HOD}_c$ .

**Theorem 2.8.** Let  $c$  be a set such that  $c$  is ordinal definable from elements of  $c$  then  $\text{HOD}_c \models \text{ZF}$ .

*Proof.* Same as before except (Pow), let  $a \in \text{HOD}_c$  and  $A = \{x \subseteq a : x \in \text{OD}_c\}$ . As  $c \in \text{OD}_c$ , so  $A \in \text{OD}_c$ . Hence  $A \in \text{HOD}_c$ . (This proof is extremely brief, actually we used Reflection Principle: there is some ordinal  $\alpha$  such that for any  $y \in \mathcal{P}(a) \cap \text{OD}_c \iff$  there is a formula  $\varphi$ , ordinal  $\alpha_0, \dots, \alpha_{n-1}$  and  $y_0, \dots, y_{m-1}$  elements from  $c$  such that  $\forall u (u \in y) \leftrightarrow V_\alpha \models \varphi(u, \alpha_0, \dots, \alpha_{n-1}, y_0, \dots, y_{m-1})$ . Since  $c \in \text{OD}_c$  so  $\mathcal{P}(a) \cap \text{OD}_c \in \text{OD}_c$ . And it is easy to verify  $A \in \text{HOD}_c$ .) ■

Now let us construct counterexamples of Axiom of Choice in  $\text{HOD}_c$ .

**Recall:**  $\mathbb{C}$  is the set of all finite sequence of natural numbers, ordered by end-extension, and the finite support product of  $\alpha$  copies of  $\mathbb{C}$ , i.e.,  $\mathbb{C}(\alpha) = \{p \mid p : \alpha \rightarrow \overline{\mathbb{C}} \wedge \overline{\{i < \alpha \mid p(i) \neq 1_{\mathbb{C}}\}} < \aleph_0\}$ . For  $p \in {}^\alpha \mathbb{C}$ , we write  $\text{supp}(p) = \{i < \alpha : p(i) \neq 1_{\mathbb{C}}\}$ , then  $p \in \mathbb{C}(\alpha)$  if and only if  $p \in {}^\alpha \mathbb{C}$  and  $\overline{\text{supp}(p)} < \aleph_0$ . From last semester we know both  $\mathbb{C}$  and  $\mathbb{C}(\alpha)$  have the c.c.c.

**Definition 2.9.** Let  $\mathbb{P}$  be a poset, we call  $\mathbb{P}$  **homogeneous** if for all  $p, q \in \mathbb{P}$  there is an isomorphism (or dense endomorphism)  $\pi$  such that  $\pi(p) \parallel q$ .

**Lemma 2.10.**  $\mathbb{C}$  is homogeneous.

*Proof.* Let  $p, q \in \mathbb{C}$  be given, define

$$\pi(r) = \begin{cases} r, & \text{if none of the following is true: } r \subseteq p, p \subseteq r, r \subseteq q, q \subseteq r \\ q \restriction \text{dom}(r), & \text{if } r \subseteq p \\ p \restriction \text{dom}(r), & \text{if } r \subseteq q \\ q \cup (r \restriction [\text{dom}(q), \text{dom}(r)]), & \text{if } p \subseteq r \\ p \cup (r \restriction [\text{dom}(p), \text{dom}(r)]), & \text{if } q \subseteq r \end{cases}$$

Then  $\pi$  is what we want. ■

**Lemma 2.11.**  $\mathbb{C}(\alpha)$  is homogeneous for all  $\alpha$ .

*Proof.* Do the same thing on  $\xi \in \text{supp}(p) \cap \text{supp}(q)$ . ■

**Lemma 2.12.** Let  $M$  be a transitive model of ZFC and  $\mathbb{P} \in M$  be a poset such that  $M \models \text{"}\mathbb{P}\text{ is homogeneous"}$ .

Let  $\pi \in M, \pi : \mathbb{P} \rightarrow \mathbb{P}$  be an isomorphism, then  $\pi$  induces a map  $\tilde{\pi} : M^{\mathbb{P}} \rightarrow M^{\mathbb{P}}$  (recursively) defined by  $\tilde{\pi}(\tau) = \{(\tilde{\pi}(\sigma), \pi(p)) : (\sigma, p) \in \tau\}$  where  $\tau \in M^{\mathbb{P}}$  with the following property: for every formula  $\varphi$  and for all  $p \in \mathbb{P}$  and for all  $\tau_1, \dots, \tau_n \in M^{\mathbb{P}}, p \Vdash \varphi(\tau_1, \dots, \tau_n) \iff \pi(p) \Vdash \varphi(\tilde{\pi}(\tau_1), \dots, \tilde{\pi}(\tau_n))$ .



*Proof.* We may assume  $M$  to be countable.

Claim: let  $g \subseteq \mathbb{P}$  be  $M$ -generic and  $h := \pi''g \subseteq \mathbb{P}$ , then  $h$  is also  $M$ -generic and  $\tau^g = \tilde{\pi}(\tau)^h$ . (If  $\pi$  is dense endomorphism then let  $h := \pi''g \uparrow$ .)

Proof of the claim:  $x \in \tau^g$  if and only if there is some  $p \in g$  and some  $\sigma$  with  $(\sigma, p) \in \tau$ . And  $x \in \tilde{\pi}(\tau)^h$  if and only if there is some  $\bar{p} \in h$  and some  $\bar{\sigma}$  with  $(\bar{\sigma}, \bar{p}) \in \tilde{\pi}(\tau)$ . Take  $\bar{p} = \pi(p)$ ,  $\bar{\sigma} = \tilde{\pi}(\sigma)$ .

And same argument shows the claim: let  $g \subseteq \mathbb{P}$  be  $M$ -generic and  $h = \pi^{-1''}g$  then  $h$  is also  $M$ -generic and  $\tau^h = \tilde{\pi}(\tau)^g$ .

Now we show  $\Rightarrow$ , let  $g \subseteq \mathbb{P}$  be any  $M$ -generic with  $\pi(p) \in g$ , let  $h = \pi^{-1''}g$  then  $p \in h$ . And thus  $M[h] \models \varphi(\tau_1^h, \dots, \tau_n^h)$ , since  $M[g] = M[h]$ , we have  $M[g] \models \varphi(\tilde{\pi}(\tau_1)^g, \dots, \tilde{\pi}(\tau_n)^g)$  thus  $\pi(p) \Vdash \varphi(\tilde{\pi}(\tau_1), \dots, \tilde{\pi}(\tau_n))$ . Converse holds by the same argument.  $\blacksquare$

**Theorem 2.13.** *Let  $M$  be a transitive model of ZFC and  $g \subseteq \mathbb{C}$  be  $M$ -generic. Then  $\bigcup g \in {}^\omega\omega$  is not definable over  $M[g]$  from any element of  $M$ .*

*Proof.* Assume there is some formula  $\varphi$  and some  $x_1, \dots, x_n \in M$  such that  $(\bigcup g)(n) = m$  if and only if  $M[g] \models \varphi(n, m, x_1, \dots, x_n)$  for all  $n, m < \omega$ . Let  $\pi: \mathbb{C} \rightarrow \mathbb{C}$  be any isomorphism,  $\pi \in M$ . For all  $x \in M$ ,  $\tilde{x} = \tilde{\pi}(x)$ . Hence if  $p, q \in \mathbb{C}$  and  $\pi \in M$  is an isomorphism with  $\pi(p) \parallel q$  then

$$p \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k) \iff \pi(p) \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$$

If  $p \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$ ,  $q \Vdash \neg \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$  but  $\pi(p) \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$  contradict to  $\pi(p) \parallel q$ . So there are no  $p, q$  with  $p \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$ ,  $q \Vdash \neg \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$ , hence  $\mathbb{1}_{\mathbb{C}} \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$  or  $\mathbb{1}_{\mathbb{C}} \Vdash \neg \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$ .

Therefore  $(\bigcup g)(n) = m \iff \exists p \in g, p \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k) \iff \mathbb{1}_{\mathbb{C}} \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$ , then  $\bigcup g \in M$  which contradict to the fact that  $\mathbb{C}$  is atomless.  $\blacksquare$

A similar argument shows the following:

**Theorem 2.14.** *Let  $M$  be a transitive model of ZFC and  $\alpha \in M, g \subseteq \mathbb{C}(\alpha)$  be a  $M$ -generic filter.*

*Let  $\chi_\xi = \bigcup \{p(\xi) : p \in g\} = \bigcup g_\xi \in {}^\omega\omega$  for  $\xi < \alpha$ , then  $\chi_\xi \neq \chi_{\xi'}$  for any  $\xi \neq \xi'$ .*

*Proof.* Suppose not, let  $\xi \neq \xi'$  be such that  $M[g] \models \chi_\xi = \chi_{\xi'}$ . By forcing theorem, there exists  $p \in g$  such that  $p \Vdash \dot{\chi}_\xi = \dot{\chi}_{\xi'}$ . But  $\mathbb{C}(\alpha)$  is homogeneous, so for any  $p, q \in \mathbb{C}(\alpha)$ , there exists an isomorphism  $\pi$  such that  $\pi(p) \parallel q$ , then by the similar argument,  $1 \Vdash \chi_\xi(\tilde{n}) = \chi_{\xi'}(\tilde{n})$  or  $1 \Vdash \chi_\xi(\tilde{n}) \neq \chi_{\xi'}(\tilde{n})$ . By assumption we have  $1 \Vdash \chi_\xi(\tilde{n}) = \chi_{\xi'}(\tilde{n})$ .

However, we can easily construct a condition  $q \leq 1$  that forces them to be different. Define a condition  $q \in \mathbb{C}(\alpha)$  such that:  $q(\gamma) = \emptyset$  for  $\gamma \neq \xi, \xi', q(\xi) = \{(n, 0)\}$ ,  $q(\xi') = \{(n, 1)\}$ . Clearly  $q \leq 1$ . But  $q \Vdash \dot{\chi}_\xi(\tilde{n}) = 0$  and  $q \Vdash \dot{\chi}_{\xi'}(\tilde{n}) = 1$ , so:

$$q \Vdash \dot{\chi}_\xi \neq \dot{\chi}_{\xi'}$$

This contradicts the fact that  $1 \Vdash \dot{\chi}_\xi = \dot{\chi}_{\xi'}$ .  $\blacksquare$

**Lemma 2.15** (Product Lemma). *Let  $M$  be a transitive model of ZFC,  $\mathbb{P}, \mathbb{Q}$  be forcing posets in  $M$ .*

(1) If  $G \subseteq \mathbb{P}$  is  $\mathbb{P}$ -generic over  $M$ ,  $H \subseteq \mathbb{Q}$  is  $\mathbb{Q}$ -generic over  $M[G]$  then  $G \times H$  is a  $\mathbb{P} \times \mathbb{Q}$ -generic filter over  $M$ .

(2) If  $K \subseteq \mathbb{P} \times \mathbb{Q}$  is  $\mathbb{P} \times \mathbb{Q}$ -generic over  $M$ , set  $G := \{p \in \mathbb{P} : \exists q \in \mathbb{Q}, (p, q) \in K\}$  and  $H := \{q \in \mathbb{Q} : \exists p \in \mathbb{P}, (p, q) \in K\}$  then  $G$  is  $\mathbb{P}$ -generic over  $M$ ,  $H$  is  $\mathbb{Q}$ -generic over  $M[G]$ .

*Proof.* (1) Let  $D \subset \mathbb{P} \times \mathbb{Q}, D \in M$  be dense. Define  $D' = \{q \in \mathbb{Q} : \exists p \in G, (p, q) \in D\}$ , clearly  $D' \in M[G]$ . We claim that  $D'$  is dense in  $\mathbb{Q}$ : Fix  $r \in \mathbb{Q}$ , we want to find  $q \in D', q \leq r$ . Define  $E_r = \{p \in \mathbb{P} : \exists q \in \mathbb{Q}, q \leq r, (p, q) \in D\}$ . For any  $s \in \mathbb{P}, (s, r) \in \mathbb{P} \times \mathbb{Q}$  thus  $\exists (p, q) \in D, (p, q) \leq (s, r)$  thus  $p \leq s$ , so  $E_r$  is dense in  $\mathbb{P}$ . Say  $p \in G \cap E_r$  then there exists  $q \in \mathbb{Q}, q \leq r, (p, q) \in D$ . Hence  $D'$  is dense in  $\mathbb{Q}$ . And then  $H \cap D' \neq \emptyset$ , gives  $G \times H \cap D \neq \emptyset$ .

(2) Let  $D \subset \mathbb{P}, D \in M$  be dense. Take  $D \times \mathbb{Q}$ , this is clearly dense in  $\mathbb{P} \times \mathbb{Q}$  thus  $G \cap D \neq \emptyset$ .

Now let  $D \subset \mathbb{Q}, D \in M[G]$  be dense. Let  $\tau^G = D$  and there is  $p^* \in G, p^* \Vdash \tau$  is dense in  $\check{\mathbb{Q}}$ . Define  $D^* = \{(p, q) \in \mathbb{P} \times \mathbb{Q} : p \leq p^* \wedge p \Vdash \check{q} \in \tau\}$ . We claim that  $D^*$  is dense below  $(p^*, 1_{\mathbb{Q}})$ .

Take any  $(p, q) \leq (p^*, 1_{\mathbb{Q}})$  since  $p \Vdash \tau$  is dense in  $\check{\mathbb{Q}}$  thus  $p \Vdash \exists \sigma \in \check{\mathbb{Q}} (\sigma \in \tau) \wedge (\sigma \leq \check{q})$ . Then by forcing theorem the set  $\{s \leq p : s \Vdash \exists t \in \mathbb{Q} (\check{t} \in \tau) \wedge (\check{t} \leq \check{q})\}$  is dense below  $p$ . Thus there exists  $s \leq p, t \in \mathbb{Q}$  such that  $s \Vdash \check{t} \in \check{\mathbb{Q}} \wedge \check{t} \leq \check{q}$  and thus  $(s, t) \in D^*$ . ■

**Warning:** If  $g$  is  $\mathbb{P}$ -generic over  $M$  and  $h$  is  $\mathbb{Q}$ -generic over  $M$  then  $g \times h$  is not necessarily a  $\mathbb{P} \times \mathbb{Q}$ -generic filter over  $M$ . See Sheet 6.

By using the product lemma, in fact the following is true:

$$\mathbb{C}(\alpha) \cong \begin{cases} \mathbb{C}(\alpha) \times \mathbb{C} & \text{if } \alpha \geq \omega \\ \mathbb{C}(\alpha - 1) \times \mathbb{C} & \text{if } \alpha < \omega \end{cases}$$

and then  $g_{\xi}$  is  $\mathbb{C}$ -generic over  $M[(g_{\gamma} : \gamma < \alpha, \gamma \neq \xi)], M[g] = M[(g_{\gamma} : \gamma < \alpha, \gamma \neq \xi)][g_{\xi}]$ , then  $\chi_{\xi}$  is not definable over  $M[g]$  from parameters in  $M[(g_{\gamma} : \gamma < \alpha, \gamma \neq \xi)]$  by the previous lemma.

**Theorem 2.16.** Let  $M$  be a transitive model of ZFC, let  $\alpha \in M, g$  be a  $\mathbb{C}(\alpha)$ -generic filter over  $M$  and  $x \in {}^{\omega}\omega \cap M[g]$ . There is then some  $h \in M[g]$ , here  $h$  is  $\mathbb{C}$ -generic over  $M$  with  $x \in M[h]$ .

*Proof.* Let  $x = \tau^g$  for every pair  $(n, m) \in \omega \times \omega$ , let  $A_{n,m}$  be an maximal antichain of  $p \in \mathbb{C}(\alpha)$  with  $p \Vdash \tau(\check{n}) = \check{m}$ , since  $\mathbb{C}(\alpha)$  has the c.c.c.  $A_{n,m}$  is countable. Let  $\bar{\tau} = \{((n, \check{m}), p) : p \in A_{n,m} ; n, m < \omega\}$  then  $\bar{\tau}^g = \tau^g$ . Let  $X := \bigcup \{\text{supp}(p) : \exists (n, m) \in \omega \times \omega \text{ with } p \in A_{n,m}\}, \bar{\bar{X}} \leq \omega$ . Since  $\mathbb{C}(\alpha) \cong \mathbb{C}(X) \times \mathbb{C}(\alpha \setminus X)$ , but  $\mathbb{C}(X) \cong \mathbb{C}(\omega) \cong \mathbb{C}$ . Here  $\mathbb{C}(X) = \{p \in {}^X\mathbb{C} : \overline{\text{supp}(p)} < \aleph_0\}$  By the choice of  $X, x = \bar{\tau}^g \in M[g \restriction_X]$  therefore  $x \in M[h]$  where  $h \in M[g]$  is  $\mathbb{C}$ -generic, i.e., we rewrite  $g \restriction_X$  as  $h$  via an isomorphism  $\pi : \mathbb{C}(X) \rightarrow \mathbb{C}$ . ■

**Theorem 2.17.** Let  $M$  be a transitive model of ZFC and  $\alpha \in M, \alpha \geq \omega_1^M$  and let  $g$  be  $\mathbb{C}(\alpha)$ -generic over  $M$ . Then in  $M[g]$  there is no well-order of the reals which are definable from ordinals and reals, i.e., there is no formula  $\varphi$  and no ordinals  $\alpha_0, \dots, \alpha_{n-1} \in M[g]$  and reals  $x_0, \dots, x_{m-1} \in M[g]$  such that  $\{(x, y) \in ({}^{\omega}\omega \times {}^{\omega}\omega) \cap M[g] : M[g] \models \varphi(x, y, \alpha_0, \dots, \alpha_{n-1}, x_0, \dots, x_{m-1})\}$  is a well-order of  ${}^{\omega}\omega \cap M[g]$ .

*Proof.* Suppose there exists a formula  $\varphi$  and ordinals  $\alpha_0, \dots, \alpha_{n-1} \in M[g]$  and reals  $x_0, \dots, x_{m-1} \in M[g]$  such that  $\{(x, y) \in (\omega^\omega \times \omega^\omega) \cap M[g] : M[g] \models \varphi(x, y, \alpha_0, \dots, \alpha_{n-1}, x_0, \dots, x_{m-1})\}$  is a well-order of  $\omega^\omega \cap M[g]$ . By what we proved earlier, we may pick a (large enough) set  $X \subseteq \alpha, X \in M$  which is countable in  $M$  such that  $x_0, \dots, x_{m-1} \in M[g \upharpoonright_X] \subseteq M[g] = M[g \upharpoonright_X][g \upharpoonright_{(\alpha \setminus X)}], \mathbb{C}(X) \times \mathbb{C}(\alpha \setminus X) \cong \mathbb{C}(\alpha)$ . Let  $\beta \in \alpha \setminus X$  and look at  $g \upharpoonright_{\{\beta\}}$  which may be constructed as a Cohen real. Cohen real over  $M[g \upharpoonright_X]$  in particular  $\bigcup g \upharpoonright_{\{\beta\}}$  is not definable in  $M[g]$  from parameters in  $M[g \upharpoonright_X]$ . By the hypothesis there is some ordinal  $\gamma \in M$  such that from the point of view of  $M[g], \bigcup g \upharpoonright_{\{\beta\}}$  is the  $\gamma^{\text{th}}$ -real in the well-order defined by  $\varphi(-, -, \alpha_0, \dots, \alpha_{n-1}, x_0, \dots, x_{m-1})$ . In particular  $\bigcup g \upharpoonright_{\{\beta\}}$  would be definable in  $M[g]$  from parameters in  $M[g \upharpoonright_X]$ , contradiction! ■

**Corollary 2.18.** *Let  $M$  be a transitive model of ZFC and  $\alpha \geq \omega_1^M, g$  be a  $\mathbb{C}(\alpha)$ -generic filter over  $M$ . Let  $\mathbb{R}^* = \omega^\omega \cap M[g]$  and  $N = (\text{HOD}_{\mathbb{R}^*})^{M[g]}$ . Then  $N \models \text{"ZF and there is no well-order of the reals"}$ .*

*Proof.* If  $N$  had a well-order, say  $\leq$ , then  $\leq$  would be definable inside  $M[g]$  using finitely many ordinals and reals from  $\mathbb{R}^*$ , contradiction! ■

### 2.1.2 A model of ZF in which there is a countable set of countable sets of reals whose union is uncountable

**Definition 2.19.** *Let  $X$  be a set and let  $\alpha$  be an ordinal. Define  $\text{Col}(\alpha, X) := \{p : \exists \beta < \alpha, p \in {}^\beta X\}$ , ordered by  $p \leq q$  if and only if  $p \supseteq q$ . To force CH, use  $\text{Col}(\omega_1, 2^{\aleph_0})$ .*

From now, we are going to make use only of  $\text{Col}(\omega, \gamma)$  for some ordinals  $\gamma$ .

**Definition 2.20.** *Let  $\beta$  be an ordinal, then*

$$\text{Col}(\omega, < \beta) := \{p : (p \text{ is a function with } \text{dom}(p) = \beta) \wedge (\forall \gamma < \beta, p(\gamma) \in \text{Col}(\omega, \gamma)) \wedge \overline{\overline{\text{supp}(p)}} < \aleph_0\}$$

*ordered by  $p \leq q \iff p(\gamma) \leq_{\text{Col}(\omega, \gamma)} q(\gamma)$  for all  $\gamma$ .*

**Lemma 2.21.** *Let  $M$  be a transitive model of ZFC and let  $X \in M$ , if  $g$  is  $\text{Col}(\omega, X)$ -generic over  $M$  then  $X$  is at most countable in  $M[g]$ .*

*Proof.* For any  $x \in X$ , define  $D_x := \{p \in \text{Col}(\omega, X) : \exists n < \omega, p(n) = x\}$ ,  $D_x \in M$  and  $D_x$  is dense, hence  $D_x \cap g \neq \emptyset$ . Let  $f = \bigcup g \in M[g]$ ,  $f$  is a function from  $\omega$  to  $X$  and by the proceeding paragraph  $f$  is actually onto. ■

**Definition 2.22.** *Let  $\kappa$  be a cardinal, then*

$$\text{Col}(\omega, < \kappa) := \{p : (p \text{ is a function with } \text{dom}(p) = \kappa) \wedge (\forall \xi < \kappa, p(\xi) \in \text{Col}(\omega, \xi)) \wedge \overline{\overline{\text{supp}(p)}} < \aleph_0\}$$

*ordered by  $p \leq q \iff p(\xi) \leq_{\text{Col}(\omega, \xi)} q(\xi)$  for all  $\xi < \kappa$ .*

Notice that  $\overline{\overline{\text{Col}(\omega, < \kappa)}} = \kappa$ .

**Lemma 2.23.** *Let  $M$  be a transitive model of ZFC and let  $\kappa \in M$ , if  $g$  is  $\text{Col}(\omega, < \kappa)$ -generic over  $M$  then  $M[g] \models \overline{\overline{\xi}} \leq \aleph_0$  for all  $\xi < \kappa$ .*

*Proof.* Directly follows from the previous lemma and the definition. ■

**Definition 2.24.**  $\text{Col}^*(\omega, Y) = \{p: (p \text{ is a function with } \text{dom}(p) = Y) \wedge (\forall X \in Y, p(X) \in \text{Col}(\omega, X)) \wedge (\overline{\text{supp}(p)} < \aleph_0)\}$ , ordered as before.

*E.g.* for all  $Y \subseteq \kappa, Y \neq \emptyset, \text{Col}(\omega, < \kappa) \cong \text{Col}^*(\omega, Y) \times \text{Col}^*(\omega, \kappa \setminus Y)$ . In particular, this gives a proof of the last lemma immediately:  $\text{Col}(\omega, < \kappa) \cong \text{Col}^*(\omega, \{\xi\}) \times \text{Col}^*(\omega, \kappa \setminus \{\xi\}), \text{Col}(\omega, \xi) = \text{Col}^*(\omega, \{\xi\})$ .

Note that if  $g$  is a  $\text{Col}(\omega, < \kappa)$ -generic filter over  $M$  then  $M[g] \models \bar{\kappa} \leq \aleph_1$ , additionally if  $\kappa$  is singular then  $M[g] \models \bar{\kappa} = \aleph_0$  as  $\text{cf}(\kappa) = \omega$  in  $M[g]$ , follows from 2.23.

Later we want to study  $\text{Col}(\omega, \kappa)$  for regular limit uncountable cardinal (they are called weakly inaccessible cardinal), for now we want to study what happens if we force with  $\text{Col}(\omega, < \aleph_\omega)$ .

**Lemma 2.25.** *Let  $M$  be a transitive model of ZFC and let  $\mathbb{P} \in M$  be a poset such that  $M \Vdash \text{"}\mathbb{P} \text{ has the } \kappa\text{-c.c."}$  for some  $\kappa \in M$  which is an uncountable cardinal of  $M$ . If  $g$  is  $\mathbb{P}$ -generic over  $M$  then  $M[g]$  and  $M$  has the same cardinal  $\geq \kappa$ .*

*Proof.* Suppose not then let  $\lambda \geq \kappa$  be the least such that  $M \models \text{"}\lambda \text{ is a cardinal"}$  and  $M[g] \models \text{"}\lambda \text{ is not a cardinal"}$ . Then  $\lambda = (\eta^+)^M$  for some  $\eta < \lambda, M[g] \models \exists f (f: \eta \rightarrow \lambda \text{ is onto})$ . Say  $f = \tau^g$  where  $\tau \in M^{\mathbb{P}}$ .

Let  $p \in g, p \Vdash \text{"}\tau: \check{\eta} \rightarrow \check{\lambda} \text{ is onto."}$  For every  $\xi < \eta$ , let  $X_\xi = \{\nu < \lambda: \exists q \leq p (q \Vdash \tau(\check{\xi}) = \check{\nu})\}$ . For each  $\nu \in X_\xi$ , pick  $q_\nu \leq p$  such that  $q_\nu \Vdash \tau(\check{\xi}) = \check{\nu}$ . Then  $\{q_\nu: \nu \in X_\xi\}$  is an antichain as  $\nu \neq \nu', q_\nu \perp q_{\nu'}$ . Hence  $\overline{X_\xi} = \overline{\{q_\nu: \nu \in X_\xi\}} < \kappa$ , i.e.,  $M \models \overline{X_\xi} < \kappa$ .

**Case 1:**  $\kappa \leq \eta$

We define  $F: \eta \times \kappa \rightarrow \lambda$  in  $M$  as follows:  $\forall \xi < \eta$ , let  $f_\xi: \kappa \rightarrow X_\xi$  be onto, for  $(\xi, i) \in \eta \times \kappa$ , let  $F((\xi, i)) = f_\xi(i)$ . We claim  $F$  is onto: Let  $\nu < \lambda$ , there is some  $\xi < \eta$  such that  $f(\xi) = \nu$  since  $f$  is onto, then  $\nu \in X_\xi$  as there is some  $q \leq p, q \in g$  with  $q \Vdash \tau(\check{\xi}) = \check{\nu}$ . Hence  $\nu = f_\xi(i)$  for some  $i < \kappa$ . Therefore  $\nu \in \text{ran}(F)$ , but then  $F \in M, \lambda \leq \overline{\eta \times \kappa}$  and contradict to the fact  $\kappa \leq \eta < \lambda$ .

**Case 2:**  $\kappa = \lambda = \eta^+$

In  $M, \kappa = \lambda = \eta^+$ , so  $\kappa$  is regular. Since  $f: \eta \rightarrow \lambda = \kappa$  is onto so it is also cofinal function.

Define in  $M, h: \eta \rightarrow \kappa$  by  $h(\xi) = \sup X_\xi$ . By the regularity of  $\kappa$ ,  $h(\xi) < \kappa$  for all  $\xi < \eta$ , so  $h$  is well-defined. Now we want to show  $h$  is a cofinal function in  $\kappa$ . Take  $\beta < \kappa$ , since  $M[g] \models \text{ran}(f)$  is cofinal in  $\kappa$ , there exists  $\xi < \eta$  such that  $f(\xi) > \beta$ . Let  $q \in g, q \leq p$  be such that  $q \Vdash \tau(\check{\xi}) > \check{\beta}$ , then  $f(\xi) \in X_\xi$ . It looks a bit weird because it is an abuse of notation. Thus  $f(\xi) < \sup X_\xi = h(\xi)$ , hence  $h(\xi) > \beta$ . So  $h \in M$  is a cofinal function but  $\eta < \text{cf}(\kappa) = \kappa$  and then leads to a contradiction. ■

**Lemma 2.26.** *Let  $M$  be a transitive model of ZFC and let  $g$  be  $\text{Col}(\omega, < \aleph_\omega^M)$ -generic over  $M$ . Then  $\aleph_{\omega+\alpha}^M = \aleph_\alpha^{M[g]}$  for all  $\alpha \geq 1$ . More precisely,  $\overline{\aleph_\omega^M} = \aleph_0^{M[g]}$  and  $\lambda$  is a cardinal in  $M[g]$  for all  $M$ -cardinals  $\lambda \geq \aleph_{\omega+1}^M$ .*

Now let us fix  $M$  to be a transitive model of ZFC and  $g \subseteq \text{Col}(\omega, < \aleph_\omega^M)$  a  $M$ -generic filter.

**Definition 2.27.** *For a cardinal  $\kappa, H_\kappa := \{x: \overline{\text{TC}(\{x\})} < \kappa\}$ , hereditarily finite  $\text{HF} = H_{\aleph_0}$ , hereditarily countable  $\text{HC} = H_{\aleph_1}$ .*

Let  $\mathbb{R}^* = \text{HC}^* = \bigcup \{ \text{HC}^{M[g \upharpoonright \alpha]} : \alpha < \aleph_\omega^M \}; \forall \alpha < \aleph_\omega^M, \text{Col}(\omega, < \aleph_\omega^M) \cong \text{Col}^*(\omega, \alpha) \times \text{Col}^*(\omega, \aleph_\omega^M \setminus \alpha)$ . By way of notation,  $g \upharpoonright_\alpha = \{p \upharpoonright_\alpha : p \in g\}$ ,  $g \upharpoonright_{(\aleph_\omega^M \setminus \alpha)} = \{p \upharpoonright_{(\aleph_\omega^M \setminus \alpha)} : p \in g\}$ , then  $g \upharpoonright_\alpha$  is  $\text{Col}(\omega, \alpha)$ -generic over  $M$  and  $g \upharpoonright_{(\aleph_\omega^M \setminus \alpha)}$  is  $\text{Col}^*(\omega, \aleph_\omega^M \setminus \alpha)$ -generic over  $M[g \upharpoonright_\alpha]$

**Proposition 2.28 (Key Claim).** *Let  $N = \text{HOD}_{\text{HC}^* \cup \{\text{HC}^*, (\aleph_n^M : n < \omega)\}}^{M[g]}$ , then  $N \models \text{ZF} \wedge \omega_1 = \aleph_\omega^M$ .*

*In particular,  $\omega_1$  has countable cofinality in  $N$  as  $(\aleph_n^M : n < \omega) \in N$ .*

We will be there if we once proved: if  $x \in \text{HC}^N$  then  $x \in \text{HC}^*$ , i.e.,  $x \in \text{HC}^{M[g \upharpoonright \alpha]}$  for some  $\alpha < \aleph_\omega^M$ . Now let us show why this statement implies  $\omega_1^N = \aleph_\omega^M$ .

**Lemma 2.29.** *Suppose  $\forall x \in \text{HC}^N, x \in \text{HC}^*$ . Then  $\omega_1^N = \aleph_\omega^M$ .*

*Proof.*  $\omega_1^N \geq \aleph_\omega^M$  : If  $\alpha < \aleph_\omega^M$ , let  $\xi = \alpha + 1$  and consider  $\text{HC}^{M[g \upharpoonright \xi]} \subset \text{HC}^* \subset N$ . Since  $\text{HC}^{M[g \upharpoonright \xi]} \models "$  $\alpha$  is countable", thus  $\aleph_\omega^M \leq \omega_1^N$ .

$\omega_1^N \leq \aleph_\omega^M$  : Suppose in  $N$  there is some  $f : \omega \rightarrow \aleph_\omega^M$  onto. Then  $f \in \text{HC}^N$  therefore by assumption  $f \in \text{HC}^*$ . Hence there is some  $\xi < \aleph_\omega^M$  with  $f \in M[g \upharpoonright \xi]$ . But since  $g \upharpoonright_\xi$  is  $\text{Col}(\omega, < \xi)$ -generic over  $M$  and  $M \models \overline{\text{Col}(\omega, < \xi)} = \bar{\xi}$  so  $\text{Col}(\omega, < \xi)$  has the  $\bar{\xi}^+$ -c.c. in  $M$  and all  $M$ -cardinals  $\geq \bar{\xi}^{+M}$  are preserved in  $M[g \upharpoonright \xi]$ . But  $\xi < \aleph_\omega^M$  so  $\aleph_\omega^M$  is still a cardinal in  $M[g \upharpoonright \xi]$  which leads to a contradiction. ■

Since  $N = (\text{HOD}_{\text{HC}^* \cup \{\text{HC}^*, (\aleph_n^M : n < \omega)\}})^{M[g]}$  so  $\text{HC}^* \subset \text{HC}^N$  holds trivially.

Now let us prove  $\forall x \in \text{HC}^N \rightarrow x \in \text{HC}^*$ .

**Lemma 2.30.** *If  $x \in \text{HC}^N$  then  $x \in \text{HC}^*$ .*

*Proof.* Let us fix  $x \in \text{HC}^N$ . Inside  $N$ , let  $f : \omega \rightarrow \text{TC}(\{x\})$  be bijective. Let  $E \subseteq \omega \times \omega$  be such that  $(n, m) \in E$  if and only if  $f(n) \in f(m)$ , i.e.,  $(\omega; E) \cong (\text{TC}(\{x\}); \in)$ . It suffices to prove  $E \in \text{HC}^*$  because then if  $E \in \text{HC}^{M[g \upharpoonright \xi]}, \xi < \aleph_\omega^M$  and then  $f, \text{TC}(\{x\}) \in M[g \upharpoonright \xi]$  (by Axiom Schema of Replacement and Separation), hence  $x \in \text{HC}^{M[g \upharpoonright \xi]} \subset \text{HC}^*$ .

Let us prove  $E \in \text{HC}^*$ . Since  $E \in N$  so  $(n, m) \in E$  if and only if  $M[g] \models \varphi(n, m, x_1, \dots, x_k, \text{HC}^*, (\aleph_n^M : n < \omega), \alpha_1, \dots, \alpha_l)$  for some formula  $\varphi$ , some elements  $x_1, \dots, x_k \in \text{HC}^*$  and ordinals  $\alpha_1, \dots, \alpha_l$ . Let us pick  $\xi < \aleph_\omega^M$  such that  $x_1, \dots, x_k \in \text{HC}^{M[g \upharpoonright \xi]}$ . Then

$$(n, m) \in E \iff M[g \upharpoonright \xi][g \upharpoonright_{[\xi, \aleph_\omega^M]}] \models \varphi(n, m, x_1, \dots, x_k, \text{HC}^*, (\aleph_n^M : n < \omega), \alpha_1, \dots, \alpha_l)$$

Notice that  $\text{HC}^*$  is the only parameter which does not lie in  $M[g \upharpoonright \xi]$ .

In fact,  $\text{Col}^*(\omega, X)$  is homogeneous for every  $X$ . Therefore if in the above definition of  $E$  the parameter  $\text{HC}^*$  does not show up then we could argue as follows:  $E \subseteq \omega \times \omega$  is definable in  $M[g] = M[g \upharpoonright \xi][g \upharpoonright_{[\xi, \aleph_\omega^M]}]$  from parameters in  $M[g \upharpoonright \xi]$ , but  $M[g]$  is a  $\text{Col}^*(\omega, [\xi, \aleph_\omega^M])$ -generic extension of  $M[g \upharpoonright \xi]$ , hence by homogeneity,  $E \in M[g \upharpoonright \xi]$ , i.e.,  $E \in \text{HC}^{M[g \upharpoonright \xi]} \subset \text{HC}^*$ .

From now on, let us assume  $\xi = 0$  for convenience.

**Claim:** There is some  $\tau \in M^{\text{Col}(\omega, < \aleph_\omega^M)}$  such that  $\tau^g = \text{HC}^*$  and for all  $p, q \in \text{Col}(\omega, < \aleph_\omega^M)$  there is an isomorphism  $\pi$  such that  $\pi(p) \parallel q$  and  $\tilde{\pi}(\tau) = \{(\tilde{\pi}(\sigma), \pi(r)) : (\sigma, r) \in \tau\} = \tau$ .

*Proof of the claim.* Let us define a name for  $\text{HC}^* \cap \mathcal{P}(\omega)$  with the above properties, i.e.,

$$\tau = \{(\sigma, p) : \exists \xi < \aleph_\omega^M (p \in \text{Col}(\omega, < \xi)) \wedge (\text{the elements of } \sigma \text{ are all pairs } (\check{n}, q), \text{ where } n < \omega, \text{supp}(q) \subset \xi)\}.$$

In order to define  $\tau$ , let us introduce the following notation: Let  $\xi < \aleph_\omega^M$ ,  $p \in \text{Col}(\omega, < \xi)$ , we write  $p^*$  for the element in  $\text{Col}(\omega, < \aleph_\omega^M)$  such that  $p^* \restriction_\xi = p$  and  $p^*(i) = \mathbb{1}_{\text{Col}(\omega, i)}$  for  $\xi \leq i < \aleph_\omega^M$ , we may also write  $\tau^* = \{(\sigma^*, p^*) : (\sigma, p) \in \tau\}$  for any  $\text{Col}(\omega, < \xi)$ -name  $\tau$  then  $\tau^{*g} = \tau^{g \restriction_\xi}$ .

Every  $x \subseteq \omega$  in  $M[g \restriction_\xi]$  has a name of the form  $\{(\check{n}, p) : n < \omega, p \in A_n\}$  where  $A_n \subseteq \text{Col}(\omega, < \xi)$ .

Take  $x = \tau^{g \restriction_\xi}$  and let  $\sigma = \{(\check{n}, p) : n < \omega \wedge p \in \text{Col}(\omega, < \xi) \wedge p \Vdash \check{n} \in \tau\}$ .

Let  $\tau = \bigcup_{\xi < \aleph_\omega^M} \tau_\xi = \bigcup_{\xi < \aleph_\omega^M} \{(\sigma, p^*) : \sigma \in M^{\text{Col}(\omega, < \xi)} \text{ is a name for a subset of } \omega \wedge p \in \text{Col}(\omega, < \xi)\}$ . For a fixed  $\xi$ ,  $\tau_\xi$  is a name for  $\mathcal{P}(\omega) \cap \text{HC}^{M[g \restriction_\xi]}$  then  $(\tau_\xi)^*$  is a name for  $\mathcal{P}(\omega) \cap \text{HC}^{M[g \restriction_\xi]}$  in  $M^{\text{Col}(\omega, < \aleph_\omega^M)}$ . Hence  $\tau^* = \bigcup_{\xi < \aleph_\omega^M} (\tau_\xi)^*$  is a name for  $\bigcup_{\xi < \aleph_\omega^M} \mathcal{P}(\omega) \cap M[g \restriction_\xi] = \mathcal{P}(\omega) \cap \text{HC}^*$ . ■

Finally, we are left with having to prove: if  $p, q \in \text{Col}(\omega, < \aleph_\omega^M)$  there is then an isomorphism  $\pi$  with  $\pi(p) \parallel q$  and  $\tilde{\pi}(\tau) = \tau$ . Let  $p, q$  be given, extend  $p, q$  in a way to  $p', q'$  such that  $\text{supp}(p') = \text{supp}(q')$  and  $\text{dom}(p'(i)) = \text{dom}(q'(i))$  for all  $i < \aleph_\omega^M$ . As for Cohen forcing, for each  $i$  there is an isomorphism  $\pi_i : \text{Col}(\omega, i) \rightarrow \text{Col}(\omega, i)$  with  $\pi_i(p'(i)) \parallel q'(i)$ . If  $i \notin \text{supp}(p')$  then put  $\pi_i = \text{id}$ , also if  $i \in \text{supp}(p')$ ,  $n \geq \text{dom}(p'(i))$  then put  $(\pi_i(r))(n) = r(n)$  for all  $r \in \text{Col}(\omega, i)$  with  $n \in \text{dom}(r)$ . For  $r \in \text{Col}(\omega, < \aleph_\omega^M)$ ,  $(\pi(r))(i) = \pi_i(r(i))$  and it is straightforward to verify  $\tilde{\pi}(\tau) = \tau$  for all  $\tau \in M^{\text{Col}(\omega, < \aleph_\omega^M)}$ . ■

In the model  $N$  which we constructed,  $\mathbb{R}$  is the countable union of countable sets of reals as  $\mathbb{R}^N = \bigcup \{\mathbb{R}^{M[g \restriction_{\aleph_n^M}]} : n < \omega\}$  and  $N \models \mathbb{R}$  is uncountable!

## 2.2 Solovay's model: A model of ZF where every set of real is Lebesgue measurable and has the Baire property and perfect set property

**Lemma 2.31.** *Let  $\kappa \geq \aleph_1$  be a regular cardinal. Then  $\text{Col}(\omega, < \kappa)$  has the  $\kappa$ -c.c.*

**Remark:** If  $\kappa$  is singular or in particular  $\aleph_\omega$  then  $\text{Col}(\omega, < \kappa)$  does not have the  $\kappa$ -c.c. as forcing with it collapse  $\kappa$ .

*Proof.* Let  $\{p_i : i < \kappa\} \subset \text{Col}(\omega, < \kappa)$ . We may assume that there is  $n_0 < \omega$  such that  $\overline{\overline{\text{supp}(p_i)}} = n_0$  for all  $i < \kappa$ , write  $\text{supp}(p_i) = \{\xi_1^i, \dots, \xi_{n_0}^i\}$  such that  $\xi_1^i < \xi_2^i < \dots < \xi_{n_0}^i$ .

**Case 1:** If there exists  $\xi < \kappa$  with  $\{i < \kappa : \xi_{n_0}^i \geq \xi\}$  has size  $< \kappa$ . Then we may assume  $\text{supp}(p_i) \subset \xi$  for all  $i < \kappa$ , then  $p_i = (p_i \restriction_\xi)^*$  (same as how we defined in the proof of 2.30). Since  $\text{Col}(\omega, < \xi)$  has cardinality  $\bar{\xi}$ , thus  $\{p_i : i < \kappa\}$  can not be an antichain in  $\kappa$ .

**Case 2:** If the set  $\{\xi_{n_0}^i : i < \kappa\}$  is unbounded in  $\kappa$ . Let  $m_0 \leq n_0$  be the least  $n$  such that  $\{\xi_{m_0}^i : i < \kappa\}$  is unbounded in  $\kappa$ . Note that if  $m_0 = 1$  then  $\{p_i : i < \kappa\}$  can not be an antichain: let  $p_1$  be any condition in the sequence. Since for all  $n, 1 \leq n \leq n_0$  the set  $\{\xi_n^i : i < \kappa\}$  is unbounded in  $\kappa$  so we can find  $p_2$  with  $\xi_1^2 > \xi_1^1$  then  $\text{supp}(p) \cap \text{supp}(q) = \emptyset$  thus  $p \parallel q$ .

Now we may assume  $m > 1$ , then there is some  $\xi < \kappa$  with  $\xi_{m_0-1}^i < \xi$  for all  $i < \kappa$  as  $\{\xi_{m_0-1}^i : i < \kappa\}$  is bounded in  $\kappa$ . Since  $\text{Col}(\omega, < \xi)$  has cardinality  $\bar{\xi}$ , then  $\overline{\overline{\{p_i \restriction_\xi : i < \kappa\}}} = \bar{\xi}$ . So  $\kappa$  many of them are pairwise equal, by regularity,  $p_i \restriction_\xi = p_j \restriction_\xi$  for all  $i, j < \kappa$ . Then by using the same argument in the case  $m_0 = 1$ , we can show that  $\{p_i : i < \kappa\}$  can not be an antichain. ■

**Remark:**  $\text{Col}(\omega, < \aleph_1) \cong \text{add } \aleph_1 \text{ many Cohen reals} = \mathbb{C}(\omega_1)$ .

**Question**(by Hausdorff): Is there an uncountable regular limit cardinal?

**Definition 2.32.** A cardinal is called **weakly inaccessible** if it is an uncountable regular limit cardinal.

An uncountable regular cardinal  $\kappa$  is called **(strongly) inaccessible** if  $\forall \mu < \kappa, 2^\mu < \kappa$ .

Since in ZFC we have  $\mu^+ \leq 2^\mu$  for all infinite cardinal  $\mu$ , so if  $\forall \mu < \kappa, 2^\mu < \kappa$  then  $\mu^+ < \kappa$  and thus  $\kappa$  is a limit cardinal. Hence all (strongly) inaccessible cardinals are also weakly inaccessible.

If GCH holds true, i.e.,  $\aleph_{i+1} = 2^{\aleph_i}$  for all  $i < \omega$  then every weakly inaccessible cardinal is also (strongly) inaccessible.

**Theorem 2.33.** Let  $\kappa$  be strongly inaccessible, then  $(V_\kappa, \epsilon) \models \text{ZFC}$ . In particular, ZFC can not prove the existence of a strongly inaccessible cardinal.

*Proof.*  $V_\kappa$  is transitive, so both (Ext) and (Fund) hold. As  $\kappa > \omega$ , (Inf) holds in  $V_\kappa$ . Since  $\kappa$  is a limit cardinal so  $V_\kappa$  is closed under  $x, y \mapsto \{x, y\}$  as well as  $x \mapsto \bigcup x$ , hence  $V_\kappa \vdash (\text{Pair}), (\text{Union})$ .

Let  $x \in V_\kappa$ , since  $\kappa$  is limit, so there exists  $\alpha < \kappa$  such that  $x \in V_\alpha$  then  $\mathcal{P}(x) \in V_{\alpha+1} \subset V_\kappa$ . Hence (Pow) holds true. If  $x \in V_\kappa$  then any subset of  $x$  also lie in  $V_\kappa$  hence  $V_\kappa \models (\text{AC})$  and (Aus).

As for (Rep), we first prove  $H_\kappa = V_\kappa$ . For  $V_\kappa \subseteq H_\kappa$ , it suffices to show that  $V_\alpha \subseteq H_\kappa$  for all  $\alpha < \kappa$ . By induction, let  $\alpha < \kappa$  be such that  $V_\beta \in H_\kappa$  for all  $\beta < \alpha$ . If  $\alpha = \beta + 1$ , say  $\overline{\overline{V_\beta}} = \mu < \kappa$  then  $\overline{\overline{V_{\beta+1}}} = 2^{\overline{\overline{V_\beta}}} = 2^\mu < \kappa$  thus  $V_\alpha \in H_\kappa$ . If  $\alpha$  is a limit, then  $V_\alpha = \bigcup_{\beta < \alpha} V_\beta$  also has size  $< \kappa$  as  $\overline{\overline{V_\beta}} < \kappa, \overline{\overline{\alpha}} < \kappa$  and  $\text{cf}(\kappa) = \kappa$ . For  $H_\kappa \subseteq V_\kappa$ , by  $\epsilon$ -induction, let  $x \in H_\kappa$  and assume inductively that  $x \subset V_\kappa$ . For any  $y \in x$ , let  $\alpha_y$  be the least ordinal such that  $y \in V_{\alpha_y}$ . Since  $\{\alpha_y : y \in x\}$  is bounded in  $\kappa$  as  $\text{TC}(\{x\})$  has cardinality  $< \kappa$ , let  $\gamma < \kappa$  be such that  $\alpha_y < \gamma$  for all  $y \in x$ . Then  $x \in V_{\gamma+1} \subset V_\kappa$  so  $x \in V_\kappa$ .

Now, let  $F: V_\kappa \rightarrow V_\kappa$  (it does not even necessarily to be definable over  $V_\kappa$ ) be a function and  $x \in V_\kappa$ . We want to show  $F''x \in V_\kappa$ . Since  $F''x$  has cardinality  $< \kappa$  and  $\overline{\overline{\text{TC}(\{F''a\})}} < \kappa$  for all  $a \in x$ . Therefore  $\text{TC}(\{F''x\}) = \{F''x\} \cup \bigcup_{a \in x} \text{TC}(\{F''a\})$  and this has size  $< \kappa$  by regularity of  $\kappa$ . And this gives (Rep) in  $V_\kappa$ . ■

**Remark:** The same argument shows that  $V_\omega = H_\omega = \text{HF} \models \text{ZFC} - (\text{Inf})$ .

We now force over a transitive model  $M$  of  $\text{ZFC} + "$  $\kappa$  is an inaccessible cardinal" with the forcing  $\text{Col}(\omega, < \kappa)$ . Let  $g$  be a  $\text{Col}(\omega, < \kappa)$ -generic filter over  $M$ , then  $\omega_1^{M[g]} = \kappa^M$ . Let us fix this situation for now.

**Lemma 2.34.** Let  $x \in {}^\omega \omega \cap M[g]$ , there is then some  $\alpha < \kappa$  such that  $x \in M[g \restriction_\alpha]$  and  $\kappa$  is inaccessible in  $M[g \restriction_\alpha]$ .

*Proof.* Instead of proving  $x \in {}^\omega \omega$ , we prove it for  $x \in \mathcal{P}(\omega)$ . As before, there is some  $\tau \in M^{\text{Col}(\omega, < \kappa)}$  such that  $\tau = \{(\tilde{n}, p) : n < \omega \wedge p \in A_n \wedge p \Vdash \tilde{n} \in \sigma\}$  where each  $A_n \subseteq \text{Col}(\omega, < \kappa)$  is an maximal antichain of  $q \in \text{Col}(\omega, < \kappa)$  deciding  $\tilde{n} \in \tau$ .

To be more precise, let  $x = \sigma^g$ , for each  $n < \omega$ , pick  $A_n$ , an maximal antichain in  $\{q : q \Vdash \tilde{n} \in \sigma \vee q \Vdash \neg(\tilde{n} \in \sigma)\}$ , i.e., the set of forcing condition who decides  $\tilde{n} \in \sigma$  and let  $\tau = \{(\tilde{n}, p) : n < \omega \wedge p \in A_n \wedge p \Vdash$

$\check{n} \in \sigma\}$ , then  $\tau^g = \sigma^g$ . For  $\sigma^g \subseteq \tau^g$ : if  $n \in x$  then there exists some  $p \in g$  such that  $p \Vdash \check{n} \in \sigma$ , since  $g \cap A_n \neq \emptyset$ , say  $\bar{p} \in g \cap A_n$ ,  $p, \bar{p}$  both lie in  $g$  so they are compatible, thus  $\bar{p} \Vdash \check{n} \in \sigma$  then  $(\check{n}, \bar{p}) \in \tau$  and  $n \in \tau^g$ . For  $\tau^g \subseteq \sigma^g$ : It is clear by checking the definition.

Since  $\text{supp}(p)$  is a finite subset of  $\kappa$  for each  $p \in \text{Col}(\omega, < \kappa)$  and  $\overline{\bigcup_{n < \omega} A_n} < \kappa$ , so there exists some  $\alpha < \kappa$  such that  $\text{supp}(p) \subset \alpha$  for all  $p \in \bigcup_{n < \omega} A_n$ .

Let  $\bar{\tau} = \{(\check{n}, p \restriction_\alpha) : n < \omega \wedge p \in A_n \wedge p \Vdash \check{n} \in \sigma\}$  then  $(\bar{\tau})^{g \restriction_\alpha} = \tau^g$ . Hence  $x \in M[g \restriction_\alpha]$ , as  $M \models \overline{\text{Col}(\omega, < \alpha)} = \bar{\alpha} < \kappa$  so  $\text{Col}(\omega, < \alpha)$  has the  $\bar{\alpha}^+$ -c.c. then  $M$  and  $M[g]$  have the same cardinal  $\geq \bar{\alpha}^+$ . It left to verify  $M[g \restriction_\alpha] \models \kappa$  is inaccessible, we need to show  $M[g \restriction_\alpha] \models \forall \mu < \kappa, 2^\mu < \kappa$ . Now fix  $\mu < \kappa$ , if  $x \in \mathcal{P}(\mu) \cap M[g \restriction_\alpha]$  then  $x = \tau^{g \restriction_\alpha}$  where  $\tau \in M, \tau = \{(\check{\xi}, p) : \xi < \mu, p \in A_\xi, p \Vdash \check{\xi} \in \sigma\}$  where  $\overline{A_\xi} \leq \bar{\alpha}$  for each  $\xi$ . Inside  $M$  there are only  $(2^\alpha)^\mu$  many such names  $\tau$  and hence in  $M[g \restriction_\alpha]$  there are only  $< \kappa$  many subsets of  $\mu$ , this shows  $M[g \restriction_\alpha] \models \forall \mu < \kappa, 2^\mu < \kappa$ . ■

The very same argument shows that if  $f \in {}^\omega \text{Ord} \cap M[g]$  then there is some  $\xi < \kappa$  with  $f \in M[g \restriction_\xi]$ .

Now let  $N = \text{HOD}_{M \cup ({}^\omega \text{Ord} \cap M[g])}^{M[g]}$ , i.e.,  $x \in N$  if and only if every element of  $\text{TC}(\{x\})$  is definable inside  $M[g]$  with parameters in  $M \cup ({}^\omega \text{Ord} \cap M[g])$ , it is clear that  $N \models \text{ZF}$ .

**Recall (Notation):** We construct  $\mathbb{R} = {}^\omega \omega$  as a metric space as follows:

For any  $x, y \in \mathbb{R}, x \neq y$ , let  $n < \omega$  be the least such that  $x(n) \neq y(n)$  then set  $d(x, y) = \frac{1}{n+1}$ .

For  $s \in {}^{<\omega} \omega$ , define  $U_s = \{x \in {}^\omega \omega : s \subset x\}$  and recursively define  $\mu(U_s)$  as follows: let  $\mu(U_{\{\emptyset\}}) = 1$ , now suppose  $\mu(U_s)$  is already defined, then let  $\mu(U_{s \smallfrown n}) = \frac{1}{2^{n+1}} \cdot \mu(U_s)$ . Note that  $U_s$  is the disjoint union of  $\{U_{s \smallfrown n} : n < \omega\}$ , we call  $U_s, s \in {}^{<\omega} \omega$  the collection of basic open sets.

A set  $\mathcal{O} \subseteq {}^\omega \omega$  is **open** if and only if  $\mathcal{O} = \bigcup_{s \in X} U_s$  for some  $X \subseteq {}^{<\omega} \omega$  (possibly  $X = \emptyset$ ) where  $\forall s_1, s_2 \in X (s_1 \neq s_2 \rightarrow U_{s_1} \cap U_{s_2} = \emptyset)$ . With this notion,  ${}^\omega \omega$  is a topological space which in fact is the topology derived from the metric we defined earlier. And we may then set  $\mu(\mathcal{O}) = \sum_{s \in X} \mu(U_s)$ .

A set  $N \subseteq {}^\omega \omega$  is called **null** if and only if  $\forall \varepsilon > 0, \exists \mathcal{O} \subseteq {}^\omega \omega$  open,  $\mathcal{O} \supseteq N, \mu(\mathcal{O}) \leq \varepsilon$ .

A set  $M \subseteq {}^\omega \omega$  is called **Lebesgue measurable** if and only if  $\exists \mathcal{O} \subseteq {}^\omega \omega$  open such that  $M \Delta \mathcal{O}$  is null.

A set  $N \subseteq {}^\omega \omega$  is called **nowhere dense** if and only if for any  $s \in {}^{<\omega} \omega$ , there exists some  $t \in {}^{<\omega} \omega, t \supset s$  such that  $N \cap U_t = \emptyset$ .

A set  $M \subseteq {}^\omega \omega$  is called **meager** if and only if  $M = \bigcup_{n < \omega} N_n$  for a collection of nowhere dense sets.

We say a set  $A \subseteq {}^\omega \omega$  has the **Baire property** if and only if  $\exists \mathcal{O} \subseteq {}^\omega \omega$  open such that  $A \Delta \mathcal{O}$  is meager.

We say  $I \subseteq \mathcal{P}({}^\omega \omega)$  is a  **$\sigma$ -ideal** if and only if  $(\forall A \in I)(\forall B \subset {}^\omega \omega)(B \subseteq A \rightarrow B \in I)$  and for any  $\{A_n : n < \omega\} \subset I, \bigcup_{n < \omega} A_n \in I$ . Note that both the collections of null sets and meager sets form a  $\sigma$ -ideal.

A set  $B \subseteq {}^\omega \omega$  is called **Borel** if  $B \in \Gamma$  where  $\Gamma \subset \mathcal{P}({}^\omega \omega)$  is the smallest class such that all open subset of  ${}^\omega \omega$  in  $\Gamma$  and close under taking complement and countable union.

A set  $P \subseteq {}^\omega \omega$  is called **perfect** if and only if  $P \neq \emptyset, P$  is closed and  $\forall x \in P, x$  is an accumulation point of  $P$ , i.e.,  $\forall n < \omega, (U_{x \restriction_n} \setminus \{x\}) \cap P \neq \emptyset$ .

We say a set  $A \subseteq {}^\omega \omega$  has the **perfect set property** if and only if either  $A$  is countable or  $A$  has a perfect subset.



**Lemma 2.35.** Let  $\mathcal{L}$  be the set of all Lebesgue measurable subsets of  ${}^\omega\omega$ . Let  $\mathcal{C}$  be the set of all  $B \subseteq {}^\omega\omega$  such that there is some open set  $A \subseteq {}^\omega\omega$  with  $B \Delta A$  being meager. Both  $\mathcal{L}$  and  $\mathcal{C}$  form a  $\sigma$ -algebra containing all the open sets.

*Proof.* See Sheet 8 Exercise 2. ■

**Corollary 2.36.** Every Borel set is Lebesgue measurable and has the Baire Property.

Let  $I \subseteq \mathcal{P}({}^\omega\omega)$  be a  $\sigma$ -ideal, then we consider the following poset: let  $\mathcal{B}$  be the collection of all Borel sets, now we define an equivalence relation  $\sim$  on  $\mathcal{B}$  as follows: for  $B_1, B_2 \in \mathcal{B}$ ,  $B_1 \sim B_2 \iff B_1 \Delta B_2 \in I$ . Now let us check if  $\sim$  is an equivalence relation, for reflexivity and symmetry it is clear, as for transitivity, let  $B_1 \sim B_2, B_2 \sim B_3$  then  $B_1 \Delta B_2 \in I, B_2 \Delta B_3 \in I$ , since  $B_1 \setminus B_3 \subseteq (B_1 \setminus B_2) \cup (B_2 \setminus B_3)$  and  $B_3 \setminus B_1 \subseteq (B_3 \setminus B_2) \cup (B_2 \setminus B_1)$ . Thus  $B_1 \Delta B_3 \subseteq (B_1 \Delta B_2) \cup (B_2 \Delta B_3)$ , so  $B_1 \Delta B_3 \in I$ , i.e.,  $B_1 \sim B_3$ . Hence  $\sim$  is an equivalence relation on  $\mathcal{B}$ . Let  $[B] = \{B' \in \mathcal{B} : B \sim B'\}$  the equivalence class of  $B \in \mathcal{B}$ , we say  $[B] \leq [C]$  if and only if  $B \setminus C \in I$ .

We write  $\mathcal{B}/I$  for the collection of all equivalence classes  $[B], B \notin I$  and ordered by  $\leq$ . This order is well-defined because: Pick  $\bar{B} \in [B], \bar{C} \in [C]$ , suppose  $\bar{B} \setminus \bar{C} \in I$  then  $B \setminus C \subseteq (B \setminus \bar{B}) \cup (\bar{B} \setminus \bar{C}) \cup (\bar{C} \setminus C) \in I$  thus  $B \setminus C \in I$  so  $[B] \leq [C]$ .

We write  $\mathcal{B}/I$  for the collection of all equivalence classes  $[B]$  where  $B \notin I$  and ordered by  $\leq$ .

**Lemma 2.37.** There is a dense embedding  $j$  from Cohen forcing  $\mathbb{C}$  to  $\mathcal{B}/\text{meager}$ .

*Proof.* Let  $j(s) = [U_s]$  where  $s \in {}^{<\omega}\omega = \mathbb{C}$ ,  $j$  is clearly an embedding. As for density, let  $[B] \in \mathcal{B}/\text{meager}$ , then by definition  $B$  is not meager and since  $B$  is Borel so  $B$  has the property of Baire, thus there is a non-empty open set  $\mathcal{O}$  with  $\mathcal{O} \Delta B$  is meager. Let  $U_s \subseteq \mathcal{O}$  then  $U_s \setminus B$  is meager thus  $[U_s] \leq [B]$ . ■

**Lemma 2.38.** Both  $\mathcal{B}/\text{null}$  and  $\mathcal{B}/\text{meager}$  have the c.c.c.

*Proof.* For  $\mathcal{B}/\text{meager}$ , since  $\mathbb{C}$  has the c.c.c. and there exists a dense embedding  $j: \mathbb{C} \rightarrow \mathcal{B}/\text{meager}$ , it is easy to check that dense embedding preserves  $\kappa$ -c.c. thus  $\mathcal{B}/\text{meager}$  has the c.c.c.

Let  $\{[B_i] : i < \omega_1\}$  be an antichain in  $\mathcal{B}/\text{null}$ . If  $[B_1], [B_2]$  are incompatible, consider  $B_1 \cap B_2$ , if  $B_1 \cap B_2$  is not null then  $[B_1 \cap B_2] \leq [B_1], [B_2]$  which makes  $[B_1], [B_2]$  compatible. Thus  $B_1 \cap B_2$  must be null. Hence  $B_i \cap B_j$  is null for all  $i \neq j$ . As none of the  $B_i$  is null,  $B_i^* = B_i \setminus \bigcup_{j < i} B_j$  is not null either. But now  $B_i^* \cap B_j^* = \emptyset$  for  $i \neq j$ . We now have for each  $i < \omega_1$ , there is  $n < \omega$  with  $\mu(B_i^*) \geq \frac{1}{n+1}$ . Then there is some  $n < \omega$  and some  $X \subseteq \omega_1$  of size  $\aleph_1$  with  $\mu(B_i^*) \geq \frac{1}{n+1}$  for all  $i \in X$  which leads to a contradiction (that  $\mu({}^\omega\omega) = 1$ ). ■

Now we look at Borel sets:  $\Sigma_1^0 =$  open subsets of  $\mathbb{R}, \Pi_1^0 =$  closed subsets of  $\mathbb{R}, \Sigma_2^0 = F_\sigma =$  all countable union of closed subsets of  $\mathbb{R}, \Pi_2^0 = G_\delta =$  all countable intersection of open subsets of  $\mathbb{R}$ .

Suppose  $\alpha < \omega_1$  and all  $\Sigma_\beta^0, \Pi_\beta^0$  are defined for  $\beta < \alpha$  then:

$\Sigma_\alpha^0 =$  all countable union of elements of  $\bigcup_{\beta < \alpha} \Pi_\beta^0, \Pi_\alpha^0 =$  all complements of elements of  $\Sigma_\alpha^0 =$  all countable intersection of elements of  $\bigcup_{\beta < \alpha} \Sigma_\beta^0$ .

It is clear that  $\mathcal{B} =$  the collection of Borel sets  $= \bigcup_{\alpha < \omega_1} \Sigma_\alpha^0 = \bigcup_{\alpha < \omega_1} \Pi_\alpha^0$ .

Now let  $M$  be a transitive model of ZFC and let  $g$  be  $M$ -generic. By recursion on  $\alpha < \omega_1^M$  we define  $B \mapsto B^* = B_g$  for  $B \in (\Sigma_\alpha^0)^M$ . If  $B$  is open in  $M$ , say  $M \models B = \bigcup_{s \in X} U_s$  for some  $X \subset {}^{<\omega}\omega$  then  $B^*$  is such that  $M[g] \models B^* = \bigcup_{s \in X} U_s$ .

Note that in  $M, U_s = \{x \in {}^\omega\omega \cap M : x \supset s\}$ ; in  $M[g], U_s = \{x \in {}^\omega\omega \cap M[g] : x \supset s\}$  since the real numbers in these two sets are different, so the basic open set is different.

**Lemma 2.39.** *Let  $M$  be a transitive model of ZFC, let  $g$  be  $(\mathcal{B}/\text{null})^M$ -generic filter over  $M$ . Then there is a unique  $x = x_g \in {}^\omega\omega \cap M[g]$  such that for all  $B \in \mathcal{B}^M, x \in B^* \iff [B] \in g$ .*

*Proof.* Existence: We work in  $M[g]$  to construct  $(S_n : n < \omega)$  where  $S_n \in {}^{<\omega}\omega, \text{dom}(S_n) = n$  such that  $[U_{S_n}^M] \in g$  for all  $n < \omega$ . Let  $S_0 = \emptyset, U_\emptyset^M = [{}^\omega\omega \cap M] = \mathbb{1}_{(\mathcal{B}/\text{null})^M} \in g$ . Now suppose  $S_n$  is defined, let  $\mathcal{D} = \{[B] \in (\mathcal{B}/\text{null})^M : \exists k < \omega, [B] \leq [U_{S_n \smallfrown k}^M]\}$  then  $\mathcal{D} \in M$ .

We claim that  $\mathcal{D}$  is dense below  $[U_{S_n}^M]$  (here actually  $[U_{S_n}^M] \in g$  by our construction). Let  $[B] \leq [U_{S_n}]$ , since  $\bigsqcup_{k < \omega} B \cap U_{S_n \smallfrown k}^M = [B]$  and  $B$  is not null, the countable union of null sets is still null, thus  $\exists k < \omega$  such that  $B \cap U_{S_n \smallfrown k}^M$  is not null in  $M$ . For such  $k < \omega, [B \cap U_{S_n \smallfrown k}^M] \leq [B]$ , and thus  $\mathcal{D}$  is dense below  $[U_{S_n}]$ . Then  $g \cap \mathcal{D} \neq \emptyset$ , say  $[B] \in g \cap \mathcal{D}$ , then set  $S_{n+1} = S_n \smallfrown k$  for such  $k$ .

And we then set  $x = \bigcup S_n$ , now we verify the property that for all  $B \in \mathcal{B}^M, x \in B^* \iff [B] \in g$  by recursion on Borel hierarchy. It is clear when  $B = U_s$  for basic open sets  $U_s, s \in {}^{<\omega}\omega$ . Suppose  $x \in B^* \iff [B] \in g$ , then  $x \notin B^* \iff x \in ({}^\omega\omega^M \setminus B)^* \iff [B] \notin g \iff [{}^\omega\omega^M \setminus B] \in g$ . Suppose  $(B_n : n < \omega) \in M, B_n \in \mathcal{B}^M$  and for all  $n < \omega, x \in B_n^* \iff [B_n] \in g$ . We want to show  $x \in (\bigcap B_n)^* = \bigcap B_n^* \Rightarrow [\bigcap B_n] \in g$ .

Since  $[B_n] \in g$  and  $[B_0 \cap B_1 \cap \dots \cap B_n] \in g$  for all  $n < \omega$ . Look at  $\bigcap B_n, {}^\omega\omega^M \setminus B_1, {}^\omega\omega^M \setminus B_2, \dots$ . If  $[D] \in g, D = \bigsqcup_{n < \omega} D_n$  then at least one  $[D_n] \in g$  because  $\{[D_n] : n < \omega\} \cup \{[{}^\omega\omega^M \setminus D]\}$  is an maximal antichain in  $(\mathcal{B}/\text{null})^M$ . Let  $[B] \leq [D]$ ,  $B \cap D$  is not null then one of  $B \cap D_n$  is not null. However none of  $[{}^\omega\omega^M \setminus B_i]$  can be in  $g$ . Hence  $\bigcap B_n$  must lie in  $g$ .

For the other direction,  $[\bigcap_{n < \omega} B_n] \in g \implies [B_n] \in g$  for all  $n < \omega$  so  $x \in B_n^*$  for all  $n < \omega$ .

Uniqueness: Let  $x, x' \in {}^\omega\omega \cap M[g]$  be such that  $\forall B \in \mathcal{B}^M, x \in B^* \iff [B] \in g \iff x' \in B^*$ . Then for all  $n < \omega, U_{x|_n}^M \subseteq B$  with  $x \in (U_{x|_n}^M)^*$  thus  $[U_{x|_n}^M] \in g$  and  $x' \in (U_{x|_n}^M)^*$ . Hence  $x = x'$ .  $\blacksquare$

The unique real  $x_g$  as above for  $(\mathcal{B}/\text{null})^M$  is called a **random real** over  $M$ .

The same holds for  $\mathcal{B}/\text{meager}$ , and the unique real for  $(\mathcal{B}/\text{meager})^M$  is called a Cohen real over  $M$ .

**Lemma 2.40.** *Let  $M$  be a transitive model of ZFC and let  $x \in {}^\omega\omega$  be in a generic extension of  $M$ . TFAE:*

- (1)  $x \in {}^\omega\omega$  is a random (Cohen) real over  $M$ .
- (2)  $x \notin B^*$  for all  $B \in \mathcal{B}^M$  such that  $M \models B$  is null (meager).

*Proof.* (1) $\implies$ (2): Let  $g$  be a  $(\mathcal{B}/\text{null})^M$ -generic over  $M$  witness  $x$  is a random real. Let  $B$  be a null set in  $M$ . Since  ${}^\omega\omega^M \setminus B \in [{}^\omega\omega^M] \in g$  so  $x \in ({}^\omega\omega^M \setminus B)^* = {}^\omega\omega^{M[g]} \setminus B^*$  then  $x \notin B^*$ .

(2) $\implies$ (1): Let  $x$  satisfies (2). Let  $g := \{[B] : B \in \mathcal{B}^M \wedge x \in B^*\}$ . It suffices to show  $g$  is  $(\mathcal{B}/\text{null})^M$ -generic over  $M$ . Let  $[B] \in g$  and  $[C] \geq [B]$ . Since  $B \setminus C$  is null, by (2),  $x \notin (B \setminus C)^*$  thus  $x \in (B \cap C)^*, [C] \in g$ . Let  $[B_1], [B_2] \in g$  since  $x \in B_1^* \cap B_2^*$ , then  $x \in U_s^{M[g]} \subseteq B_1^* \cap B_2^* = (B_1 \cap B_2)^*$  and

$g \ni [B_1 \cap B_2] \leq [B_1], [B_2]$ . Now let  $A \subseteq (\mathcal{B}/\text{null})^M$  be an maximal antichain,  $A \in M, M \models A$  is countable. So  $A = \{[B_n] : n < \omega\} \in M$ . Therefore  ${}^\omega\omega \setminus \bigcup B_n$  is null, otherwise the antichain would not be maximal. Hence  $x \in \bigcup B_n^*$  so  $\exists n < \omega, x \in B_n^*$ . Thus  $g \cap A \neq \emptyset$ . ■

**Lemma 2.41.** *Let  $M$  be a transitive model of ZFC and  $g$  be a  $M$ -generic then in  $M[g]$  :*

*If  $\mathbb{R} \cap M$  is countable in  $M[g]$  then*

- (1)  $A = \{x \in {}^\omega\omega : x \text{ is not random over } M\}$  is a null set.
- (2)  $A = \{x \in {}^\omega\omega : x \text{ is not Cohen over } M\}$  is a meager set.

*Proof.* We show (1) as the similar argument gives (2).

By Lemma 2.40,  $x \in A \iff x \in B^*$  for some  $B \in M$  such that  $M \models "B \text{ is a Borel null set}"$ . If  $M \models B$  is Borel null, then  $M[g] \models B^*$  is Borel null (**Why?** Because the Lebesgue measure is absolute between  $M$  and  $M[g]$ ). Therefore  $A = \bigcup \{B^* : B \in M \wedge M \models B \text{ is Borel and null}\}$ . Since  $M \models$  there are  $2^{\aleph_0}$  many Borel sets as there are  $2^{\aleph_0}$  many open subsets. Hence in  $M[g]$ ,  $A$  is a countable union of null sets thus still null. ■

**Definition 2.42.** *Let  $M$  be a transitive model of ZFC and  $g$  be  $M$ -generic. A set  $A \in M[g]$  such that  $M[g] \models "A \text{ is a set of reals}"$  is called **Solovay over  $M$**  if and only if there is a formula  $\varphi$  and there are parameters  $x_1, \dots, x_k \in M$  such that for all  $z \in {}^\omega\omega \cap M[g]$  which are generic over  $M$ ,*

$$z \in A \iff M[z] \models \varphi(z, x_1, \dots, x_k).$$

**Remark:** Here we may define  $z$  is generic over  $M$  as:  $z$  is a random real or a Cohen real. This suffices for our purpose. It is not hard to see that every Borel set  $B^*$  where  $B \in M$  is Solovay over  $M$ . We will eventually prove that every set of reals in  $M$  is Solovay over some  $M[g \restriction \xi], \xi < \kappa$ .

**Lemma 2.43.** *Let  $M$  be a transitive model of ZFC and  $g$  be a  $M$ -generic.*

*Let  $A \in M[g]$ , a subset of  ${}^\omega\omega \cap M[g]$ , be Solovay over  $M$ . Then*

- (1) *there is some  $B \in M[g]$ ,  $B$  a Borel set in  $M[g]$  such that  $A \triangle B \subset \{x \in {}^\omega\omega \cap M[g] : x \text{ is not random over } M\}$ .*
- (2) *there is some  $B' \in M[g]$ ,  $B'$  a Borel set in  $M[g]$  such that  $A \triangle B' \subset \{x \in {}^\omega\omega \cap M[g] : x \text{ is not Cohen over } M\}$ .*

*Proof.* We show (1) as the very similar argument gives (2).

By assumption, there is a formula  $\varphi$  and some elements  $x_1, \dots, x_k \in M$  such that  $z \in A \iff M[z] \models \varphi(x_1, \dots, x_k)$  for any  $z$  which is generic over  $M$ . It suffices to verify the property above for reals  $z$  which are random over  $M$ . Let  $\tau$  be a canonical name in  $M^{(\mathcal{B}/\text{null})^M}$  for a random real over  $M$ . Let  $E \subset (\mathcal{B}/\text{null})^M, E \in M$  be an maximal antichain in  $M$  consist of  $[D]$  with  $[D] \Vdash_M^{(\mathcal{B}/\text{null})^M} \varphi(\tau, \check{x}_1, \dots, \check{x}_k)$ . Since  $\mathcal{B}/\text{null}$  has the c.c.c. so  $\overline{\overline{E}} \leq \aleph_0$  in  $M$ . Write  $E = \{[D_n] : n < \omega\}$  in  $M$ . Set  $B = \bigcup_{n < \omega} D_n^* = \left( \bigcup_{n < \omega} D_n \right)^*$ , then  $B$  is Borel in  $M[g]$ . We claim that if  $z \in M[g]$  is random over  $M$ , then  $z \in A \iff z \in B$ . Now let  $z \in M[g]$  be random over  $M$ . Then  $z \in A \iff M[z] \models \varphi(z, x_1, \dots, x_k) \iff \exists [D] \in h \cap E, [D] \Vdash_M^{(\mathcal{B}/\text{null})^M} \varphi(\tau, \check{x}_1, \dots, \check{x}_k) \iff \exists n < \omega, [D_n] \in h \iff [\bigcup D_n] \in h \iff z \in B$ .

\*: Since  $z$  is random so  $z = \tau^h$  for some  $(\mathcal{B}/\text{null})^M$ -generic  $h$ ,  $M[z] = M[\tau^h]$ . ■

**Corollary 2.44.** *Let  $M, g, M[g]$  be as before such that  $\mathbb{R} \cap M$  is countable in  $M[g]$ . If  $A \in M[g]$ , a set of reals in  $M[g]$ , is Solovay over  $M$ . Then  $A$  is lebesgue measurable and has the Baire property.*

*Proof.* Follows immediately by the two previous lemma. ■

**Definition 2.45.** *A forcing poset  $\mathbb{P}$  is called separative if and only if  $\forall p \in \mathbb{P}, \forall q \in \mathbb{P} (q \not\leq p \rightarrow (\exists r \in \mathbb{P}, r \leq q \wedge r \perp p))$ .*

**Lemma 2.46.** *Let  $\kappa$  be an infinite cardinal and  $\mathbb{P}$  be an atomless forcing poset such that  $\mathbb{1}_{\mathbb{P}} \Vdash \bar{\kappa} = \check{\aleph}_0$ . Then for every  $p \in \mathbb{P}$ , there is an antichain  $A \subseteq \{q \in \mathbb{P}: q \leq p\}$  of size  $\kappa$ .*

*Proof.* Fix  $p \in \mathbb{P}$ . First, assume that  $\kappa = \omega$ . Now we construct two sequences of conditions  $(p_n: n < \omega)$  and  $(q_n: n < \omega)$  as follows: Let  $p_0 = p$ , suppose  $p_n$  is given then let  $p_{n+1}, q_n$  be two incompatible extension of  $p_n$  then  $(q_n: n < \omega)$  is the desired sequence.

Now assume  $\text{cf}(\kappa) = \omega < \kappa$ , let  $(\kappa_n: n < \omega)$  be a sequence of uncountable regular cardinals which is cofinal of  $\kappa$ . Then construct a sequence of conditions  $(p_n: n < \omega)$  and a sequence of antichains in  $\mathbb{P}$   $(A_n: n < \omega)$  as follows:

Set  $p_0 = p$ , suppose  $p_n$  is given, notice that  $p_n \Vdash \bar{\kappa} = \check{\aleph}_0$  by lemma 6.32 in Schindler's book, there exists an antichain  $A \subseteq \{p \in \mathbb{P}: p \in p_n\}$  of size  $\kappa_n$ . This is because if  $\mathbb{P}$  has the  $\kappa_n$ -c.c. then  $\kappa_n$  will be a cardinal in the generic model as it is uncountable regular, but since  $\mathbb{1}_{\mathbb{P}} \Vdash \bar{\kappa} = \check{\aleph}_0$ , which means  $\kappa_n$  is no longer a cardinal in the generic model, contradiction. Thus there must be an antichain with size of  $\kappa_n$  in  $\{q \in \mathbb{P}: q \leq p_n\}$ . Let  $A_n$  be such an antichain and let  $p_{n+1} \in A_n$  be arbitrary. Then  $\bigcup \{A_n \setminus \{p_{n+1}\}: n < \omega\} \subset \{q \in \mathbb{P}: q \leq p\}$  is an antichain of size  $\kappa$ .

Finally we assume  $\text{cf}(\kappa) > \omega$ . Let  $r \leq p, \tau \in M^{\mathbb{P}}$  such that  $r \Vdash \check{\omega} \rightarrow \check{\kappa}$  is surjective. Let us suppose that every antichain in  $\{q \in \mathbb{P}: q \leq r\}$  have size smaller than  $\kappa$ . For every  $n < \omega$ , let  $A_n$  be an maximal antichain in  $\{q \in \mathbb{P}: q \leq r \wedge \exists \xi q \Vdash \tau(\check{n}) = \check{\xi}\}$ . By our hypothesis,  $\overline{A_n} < \kappa$  for all  $n < \omega$  so by  $\text{cf}(\kappa) > \omega$ , there can not be a surjective function  $f: (\omega \times \bigcup \{A_n: n < \omega\}) \rightarrow \kappa$ .

However we now construct such an  $f$  by setting, for  $n < \omega$  and  $p \in \bigcup \{A_n: n < \omega\}$ ,

$$f((n, p)) = \begin{cases} \xi & \text{if } p \Vdash \tau(\check{n}) = \check{\xi} \\ 0 & \text{if no } \xi < \kappa \text{ such that } p \Vdash \tau(\check{n}) = \check{\xi} \end{cases}$$

If  $\xi < \kappa$  then there is some  $r' \leq r, n < \omega$  such that  $r' \Vdash \tau(\check{n}) = \check{\xi}$ . But then  $q \parallel r'$  for some  $q \in A_n$  by the choice of  $A_n$ , so that  $q \Vdash \tau(\check{n}) = \check{\xi}$ . Therefore  $f$  is surjective, which leads to a contradiction. ■

**Lemma 2.47.** *Let  $\mathbb{P}$  be a separative forcing poset and  $\mu \geq \aleph_1$  be a cardinal, if  $\bar{\mathbb{P}} = \mu$  and  $\mathbb{1}_{\mathbb{P}} \Vdash \check{\mu}$  is countable" then there is a dense embedding  $e: \text{Col}(\omega, \mu) \rightarrow \mathbb{P}$ .*

*Proof.* Let  $g$  be a  $\mathbb{P}$ -generic filter over  $M$  and  $\tau$  be such that  $\mathbb{1}_{\mathbb{P}} \Vdash \check{\omega} \rightarrow \check{\mu}$  is onto". We construct  $e$  by recursion on the size of domain. First set  $e(\mathbb{1}_{\text{Col}(\omega, \mu)}) = \mathbb{1}_{\mathbb{P}}$ . Suppose  $p \in \text{Col}(\omega, \mu)$ ,  $\text{dom}(p) = n$  and  $e(p)$  was already chosen. Since  $\mathbb{1}_{\mathbb{P}} \Vdash \bar{\mu} \leq \aleph_0$  and  $\bar{\mathbb{P}} = \mu$ , by 2.46, we may pick some maximal antichain  $A \subseteq \mathbb{P}$  of conditions with  $\bar{A} = \mu, q \leq e(p)$  such that  $q$  decides  $\tau(\check{n})$ , i.e., there exist some  $r \in \mathbb{P}$  with  $q \Vdash \tau(\check{n}) = \check{r}$ . We write  $A = \{q_\xi: \xi < \mu\}$ , then define  $e(p \cup \{(n, \xi)\}) = q_\xi$  for  $\xi < \mu$ .

Thus for each  $n < \omega$ ,  $A_n = \{e(p) : p \in \text{Col}(\omega, \mu), \text{dom}(p) = n\}$  is an maximal antichain of conditions  $q \in \mathbb{P}$  such that  $q$  decides  $\tau(\check{0}), \dots, \tau(\check{n} - 1)$ .

To see that  $e$  is dense, let  $r \in \mathbb{P}, r \Vdash \check{r} \in \dot{g}$ . Hence there are some  $s \leq r$  such that  $s \Vdash \tau(\check{n}) = \check{r}$  for some  $n < \omega$  (**By density argument**). Let  $t \leq s$  be such that  $t$  decides  $\tau(\check{0}), \dots, \tau(\check{n})$ , then there is some  $p \in \text{Col}(\omega, \mu)$  such that  $e(p) \in A_{n+1}, e(p) \parallel t$ . Then we have  $e(p) \Vdash \tau(\check{n}) = \check{r}$ . Hence  $e(p) \Vdash \check{r} \in \dot{g}$ . As  $\mathbb{P}$  is separative, this gives  $e(p) \leq r$ , otherwise then  $\exists u \leq e(p), u \perp r$  but then  $u \Vdash \check{r} \in \dot{g}$  and  $u \parallel r$  which leads to a contradiction.  $\blacksquare$

### 2.2.1 The BIG Theorem and its LONG Proof

**Theorem 2.48** (Part 1). *Let  $\kappa$  be inaccessible and  $G \subseteq \text{Col}(\omega, < \kappa)$  be  $M$ -generic. Then in  $M[G]$ , every set of reals which is  $\text{OD}_{\omega_\omega}$  is Lebesgue measurable and has the Baire property.*

**Recall:** By Lemma 2.34, every  $x \in H_{\omega_1^{M[G]}}^M[G]$  appears in  $M[G \restriction_\lambda]$  for some  $\lambda < \kappa$ .

Now we begin to proof this theorem.

**Claim 2.49.** *Let  $\lambda < \kappa, \mathbb{P} \in H_{\kappa}^{M[G]}$  (since  $\kappa$  collapse to  $\omega_1$  in  $M[G]$  so  $H_{\kappa}^{M[G]} \subseteq H_{\omega_1^{M[G]}}^{M[G]}$ ). Let  $S \in M[G]$  be  $\mathbb{P}$ -generic over  $M[G \restriction_\lambda]$ . Then there is  $H \subseteq \text{Col}(\omega, < \kappa)$  which is generic over  $M[G \restriction_\lambda][S]$  such that  $M[G] = M[G \restriction_\lambda][S][H]$ .*

**Why we only consider the  $\text{OD}_{\omega_\omega}$  sets?**

**What will happen if this is true?** Let  $A \subseteq {}^\omega\omega$  in  $M[G]$  be  $\text{OD}_{\omega_\omega}$ , then there is a formula  $\varphi$ , some reals  $\vec{x}$  and some ordinals  $\vec{\alpha}$  such that for any real  $x, x \in A \iff M[G] \models \varphi(x, \vec{x}, \vec{\alpha})$ . Let  $\lambda < \kappa$  be such that  $\vec{x} \in M[G \restriction_\lambda]$ . By Corollary 2.44, to show  $A$  is Lebesgue measurable and has the Baire property, we need to show 2 things: First,  $(2^{\aleph_0})^{M[G]}$  is countable which holds true trivially. Second,  $A$  is Solovay over  $M[G \restriction_\lambda]$ . To show this, let  $x \in M[G] \cap {}^\omega\omega$  be generic over  $M[G \restriction_\lambda], x \in A \iff M[G] \models \varphi(x, \vec{x}, \vec{\alpha})$ , by Claim 2.49,  $\iff M[G \restriction_\lambda][x][H] \models \varphi(x, \vec{x}, \vec{\alpha}) \xleftrightarrow{\text{by FT}} \exists p \in H, p \Vdash_{\text{Col}(\omega, < \lambda)}^{M[G \restriction_\lambda][x]} \varphi(\dot{x}, \vec{x}, \vec{\alpha})$ , by homogeneity,  $\iff \mathbb{1}_{\text{Col}(\omega, < \lambda)} \Vdash \varphi(\dot{x}, \vec{x}, \vec{\alpha})$ , it is equivalent to  $M[G \restriction_\lambda][x] \models \varphi(x, \vec{x}, \vec{\alpha})$ , so  $A$  is Solovay over  $M[G \restriction_\lambda]$ .

*Proof.* For notation simplicity, we may assume  $\lambda = 0$ . By lemma 2.34, there exists some  $\mu + 1 < \kappa$  such that  $S \in M[G \restriction_{\mu+1}]$ . Note that  $\overline{\text{Col}(\omega, < \mu + 1)} = \bar{\mu}$  in  $M$  and also  $\mathbb{1}_{\text{Col}(\omega, < \mu+1)} \Vdash \check{\mu}$  is countable. By lemma 2.47, there is a dense embedding  $i: \text{Col}(\omega, \mu) \rightarrow \text{Col}(\omega, < \mu + 1)$ .

Let  $G_0 = \{p \in \text{Col}(\omega, \mu) : i(p) \in G \restriction_{\mu+1}\}, M[G_0] = M[G \restriction_{\mu+1}]$ . Also  $j: \text{Col}(\omega, \mu) \cong \text{Col}(\omega, \{\mu + 1\})$ , let  $G_1 = \{p \in \text{Col}(\omega, \mu) : \exists q \in G, q(\mu + 1) = j(p)\}$ . Then  $G_1$  is  $\text{Col}(\omega, \mu)$ -generic over  $M[G_0] = M[G \restriction_{\mu+1}]$  such that  $M[G_0][G_1] = M[G \restriction_{\mu+2}]$ .

**Claim 2.50.** *There is some  $\text{Col}(\omega, \mu)$ -generic filter  $H^*$  over  $M[S]$  with  $M[S][H^*] = M[G_0][G_1]$ .*

**What will happen if this is true?** We will then have  $M[G] = M[G \restriction_{\mu+2}][G \restriction_{[\mu+2, \kappa)}] = M[G_0][G_1][G \restriction_{[\mu+2, \kappa)}] = M[S][H^*][G \restriction_{[\mu+2, \kappa)}]$ . Let  $i': \text{Col}(\omega, \mu) \rightarrow \text{Col}(\omega, < \mu + 2)$  be a dense embedding. Let  $H_0 = \{p \in \text{Col}(\omega, < \mu + 2) : \exists q \in H^*, i'(q) \leq p\}$ . Then  $H_0$  is  $M[S]$ -generic with  $M[S][H_0] = M[S][H^*]$ . Finally set  $H = \{p \in \text{Col}(\omega, < \kappa) : p \restriction_{\mu+2} \in H_0 \wedge p \restriction_{[\mu+2, \kappa)} \in G \restriction_{[\mu+2, \kappa)}\}$ , then  $H$  is  $\text{Col}(\omega, < \kappa)$ -generic over  $M[S]$  and  $M[S][H] = M[S][H_0][G \restriction_{[\mu+2, \kappa)}] = M[S][H^*][G \restriction_{[\mu+2, \kappa)}] = M[G]$ . And this gives claim 2.49.

*Proof.* As  $S \in M[G_0]$ , there is a  $\text{Col}(\omega, \mu)$ -name  $\tau \in M$  such that  $S = \tau^{G_0}$ .

Recursively define  $(\mathbb{Q}_\alpha : \alpha \in \text{Ord})$  of subsets of  $\text{Col}(\omega, \mu)$ . Set  $p \in \mathbb{Q}_0$  if and only if

$$\forall r \in \mathbb{P}, \begin{cases} p \Vdash \check{r} \in \tau \implies r \in S \\ p \Vdash \check{r} \notin \tau \implies r \notin S \end{cases}$$

For  $\alpha + 1, p \in \mathbb{Q}_{\alpha+1} \iff$  for any open dense set  $D \subseteq \text{Col}(\omega, \mu), D \in M, \exists p' \leq p: p' \in D \cap \mathbb{Q}_\alpha$ .

For  $\alpha$  is limit,  $\mathbb{Q}_\alpha = \bigcap_{\beta < \alpha} \mathbb{Q}_\beta$ . Note that  $\mathbb{Q}_\alpha$ s are upwards closed. And  $\alpha < \beta \implies \mathbb{Q}_\beta \subseteq \mathbb{Q}_\alpha$  because if  $p \in \mathbb{Q}_{\alpha+1}$  then  $\exists p' \leq p, p' \in D \cap \mathbb{Q}_\alpha$  so  $p \in \mathbb{Q}_\alpha$ .

It must be the case that  $\exists s < \text{Ord}$  such that  $\mathbb{Q}_\delta = \mathbb{Q}_{\delta+1}$ , i.e., the sequence will become stationary at some point and let  $\delta$  be the least such. Denote  $\overline{\mathbb{Q}} = \mathbb{Q}_\delta, \mathbb{Q} = \overline{\mathbb{Q}} \times \text{Col}(\omega, \mu)$ .

We construe  $\overline{\mathbb{Q}}$  and  $\mathbb{Q}$  as partial orders, with the order relation given by the restriction of the order relation of  $\text{Col}(\omega, \mu)$  and  $\text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu)$  to  $\overline{\mathbb{Q}}$  and  $\mathbb{Q}$  respectively. If  $p \in \overline{\mathbb{Q}}$  and  $p \leq q, q \in \text{Col}(\omega, \mu)$  then  $q \in \overline{\mathbb{Q}}$ .

Notice that  $\overline{\mathbb{Q}}$  was defined inside  $M[S]$  with the parameters  $\mu, \mathbb{P}, \tau, S$ . We write  $\Psi(\nu_0, \nu_1, \nu_2, \nu_3, \nu_4)$  for the defining formula, i.e.,  $M[S] \models \forall \mathbb{Q}' (\mathbb{Q}' = \overline{\mathbb{Q}} \leftrightarrow \Psi(\mathbb{Q}', \mu, \mathbb{P}, \tau, S))$ .

**Subclaim 2.51.**  $G_0 \subseteq \overline{\mathbb{Q}}$ .

*Proof.* Suppose  $p \in G_0 \setminus \overline{\mathbb{Q}}$  where  $\beta$  is the least such  $G_0 \setminus \overline{\mathbb{Q}}_\beta \neq \emptyset$ . By the construction of  $\overline{\mathbb{Q}}$ ,  $\beta$  can not be 0 or a limit ordinal, thus  $\beta = \alpha + 1$  for some  $\alpha$ . Since  $p \notin \overline{\mathbb{Q}}_\beta$  so there exists an open dense  $D \subseteq \text{Col}(\omega, \mu), D \in M$  such that  $\forall p' \leq p, p' \notin D \cap \mathbb{Q}_\alpha$ . Notice  $G_0$  is generic so  $G_0 \cap D \neq \emptyset$ , say  $p^* \in G_0 \cap D$ . Then  $p, p^* \in G_0$  there exist  $r \in G_0, r \leq p, p^*$ . But  $D$  is open so  $r \in D \cap \mathbb{Q}_\alpha$  hence  $G_0 \setminus \overline{\mathbb{Q}}_\alpha \neq \emptyset$ , which contradicts the choice of  $\beta$ . ■

The subclaim above gives  $\overline{\mathbb{Q}} \neq \emptyset$  also  $\overline{\mathbb{Q}}$  is separative and has size  $\mu$ .

**Subclaim 2.52.** Let  $p \in \overline{\mathbb{Q}}$ . Then in  $M[G]$  there is some  $G'_0$   $\text{Col}(\omega, \mu)$ -generic over  $M$  such that  $p \in G'_0, S = \tau^{G'_0}$  where  $\tau$  was chosen such that  $\tau^{G_0} = S$ .

*Proof.* Let  $\bar{p} \in \overline{\mathbb{Q}}, D \subset \text{Col}(\omega, \mu), D \in M$  be open dense. There is some  $p' \leq \bar{p}$  with  $p' \in D \cap \mathbb{Q}_\delta$  by  $\bar{p} \in \mathbb{Q}_{\delta+1} = \overline{\mathbb{Q}}$ . Inside  $M[G]$ , there are only countably many dense subset of  $\text{Col}(\omega, \mu)$  which are in  $M$ . Therefore inside  $M[G]$ , given any  $p \in \overline{\mathbb{Q}}$ , we may construct a  $\text{Col}(\omega, \mu)$ -generic filter  $G'_0$  over  $M$  with  $p \in G'_0, G'_0 \subset \overline{\mathbb{Q}}$ .

Let  $r \in \mathbb{P}$  and suppose that  $p' \in G'_0$  decides  $\check{r} \in \tau$ . As  $p' \in G'_0 \subset \overline{\mathbb{Q}} \subset \mathbb{Q}_0$ , hence  $p' \Vdash \check{r} \in \tau \implies r \in S, p' \Vdash \check{r} \notin \tau \implies r \notin S$ . Therefore  $G'_0$  is as desired. ■

**Subclaim 2.53.**  $G_0 \times G_1$  is  $\mathbb{Q}$ -generic over  $M[S]$ .

*Proof.* Let  $D \in M[S]$  be dense in  $\mathbb{Q}$ . We want to show  $D \cap (G_0 \times G_1) \neq \emptyset$ . Suppose  $D \cap (G_0 \times G_1) = \emptyset$ .

Let  $\rho \in M^\mathbb{P}$  be such that  $\rho^S = D$ . Let  $\Phi(\nu_0, \nu_1, \nu_2, \nu_3, \nu_4)$  be a formula such that if  $G''_0 \times G''_1$  is  $\text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu)$ -generic over  $M$ , then  $M[G''_0 \times G''_1] \models \Phi(G''_0, G''_1, \mu, \mathbb{P}, \tau)$  if and only if the following holds true: If  $s' = \tau^{G''_0}$  then  $s'$  is  $\mathbb{P}$ -generic over  $M$  and if  $D' = \{(p, q) \in \text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu) : \exists r \in$

$s' \wedge r \Vdash_M^{\mathbb{P}} (\check{p}, \check{q}) \in \rho\}$  and  $M[S] \models \Psi(\mathbb{Q}', \mu, \mathbb{P}, \tau, S)$  is true for exactly one  $\mathbb{Q}'$ , then  $D$  is dense in  $\mathbb{Q}'$  and  $D' \cap (G_0'' \times G_1'') = \emptyset$ .

By hypothesis  $M[G_0 \times G_1] \models \Phi(G_0, G_1, \mu, \mathbb{P}, \tau)$ , let  $(p, q) \in G_0 \times G_1$  be such that

$$(p, q) \Vdash_M^{\text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu)} \Phi(\dot{G}_0, \dot{G}_1, \check{\mu}, \check{\mathbb{P}}, \check{\tau})$$

where  $\dot{G}_0, \dot{G}_1 \in M^{\text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu)}$  are the canonical name for  $G_0, G_1$  respectively. By subclaim 2.51,  $(p, q) \in \mathbb{Q}$ . Since  $D$  is dense in  $\mathbb{Q}$ , there is some  $(p', q') \leq (p, q)$  such that  $(p', q') \in D$ . By subclaim 2.52, there is some  $G_0' \times G_1'$  inside  $M[G]$  which is  $\text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu)$ -generic over  $M$  such that  $(p', q') \in G_0' \times G_1'$  and  $\tau^{G_0'} = s$ .

Now we have  $M[G_0' \times G_1'] \models \Phi(G_0', G_1', \mu, \mathbb{P}, \tau)$  and  $s' = \tau^{G_0'}$  and  $D' = D$  for the  $s', D'$  which are mentioned in  $\Phi(G_0', G_1', \mu, \mathbb{P}, \tau)$  also the  $\mathbb{Q}' = \mathbb{Q}$ . We then have  $D \cap G_0' \times G_1' = \emptyset$ .

However  $(p', q') \in D \cap G_0' \times G_1'$ . **CONTRADICTION!!** ■

Now by lemma 2.47, there is a dense embedding  $k: \text{Col}(\omega, \mu) \rightarrow \mathbb{Q}$ , set  $H^* = \{p \in \text{Col}(\omega, \mu) : k(p) \in G_0 \times G_1\}$ , then  $H^*$  is  $\text{Col}(\omega, \mu)$ -generic over  $M[S]$  and  $M[S][H^*] = M[S][G_0 \times G_1] = M[G_0][G_1]$  as desired. ■

Now we finish the proof of inside  $N = \text{HOD}_{M \cup (\omega \text{Ord} \cap M[G])}^{M[G]}$ , every set of reals is Lebesgue measurable and has the Baire property. ■

Now let us show the perfect set property.

**Theorem 2.54** (Part 2). *Let  $\kappa$  be inaccessible and  $G \subseteq \text{Col}(\omega, < \kappa)$  be  $M$ -generic. Then in  $M[G]$ , every uncountable set of reals which is  $\text{OD}_{\omega_\omega}$  has a perfect subset.*

*Proof.* Fix  $A \subset N$  be an uncountable set of reals, then  $x \in A \iff M[G] \models \varphi(x, z_0, \dots, z_k, u_0, \dots, u_l)$  where  $z_0, \dots, z_k \in M, u_0, \dots, u_l \in M[G] \cap {}^\omega \omega$ . We showed that there are some  $\xi < \kappa$  such that  $u_0, \dots, u_l \in M[G \restriction_\xi]$  and  $\overline{{}^\omega \omega \cap M[G \restriction_\xi]} = \aleph_0$  in  $M[G]$ . Therefore  $A \setminus M[G \restriction_\xi] \neq \emptyset$ . Say  $x \in A \setminus M[G \restriction_\xi]$ , let  $\eta > \xi$  be such that  $x \in M[G \restriction_{(\eta+1)}]$ . There is a dense embedding  $e: \text{Col}(\omega, \eta) \rightarrow \text{Col}(\omega, [\xi, \eta])$ , let  $G^* = e^{-1} G \restriction_{[\xi, \eta]}$  then  $G^*$  is  $\text{Col}(\omega, \eta)$ -generic over  $M[G \restriction_\xi]$ ,  $x \in M[G \restriction_\xi][G^*] \setminus M[G \restriction_\xi]$ .

Let  $\tau \in M[G \restriction_\xi]^{\text{Col}(\omega, \eta)}$  be such that  $\tau^{G^*} = x$ , let us construct  $(p_s : s \in {}^{<\omega} 2)$  as follows:

Let  $(D_n : n < \omega) \in M[G]$  be an enumeration of dense subsets of  $\text{Col}(\omega, \eta)$  which exists in  $M[G \restriction_\xi]$ . Since  $M[G] = M[G \restriction_\xi][G^*][G \restriction_{[\eta+1, \kappa)}]$   $M[G] \models \varphi(x, \vec{z}, \vec{u}) \iff \mathbb{1}_{\text{Col}(\omega, [\eta+1, \kappa))} \Vdash_{M[G \restriction_\xi][G^*]} \varphi(x, \vec{z}, \vec{u})$ . This gives  $G^* \ni p_\emptyset \Vdash \mathbb{1}_{\text{Col}(\omega, [\eta+1, \kappa))} \Vdash_{M[G \restriction_\xi][G^*]} \varphi(x, \vec{z}, \vec{u}) \wedge \tau \notin M[G \restriction_\xi]$  for some  $p_\emptyset$ . Now we fix such  $p_\emptyset$  and assume  $p_s \in \text{Col}(\omega, \eta)$  was already constructed,  $p_s \leq p_\emptyset$ . There must be some  $n < \omega$  such that  $p_s$  does not decide  $\tau(\check{n})$  as  $p_\emptyset \Vdash \tau \notin M[G \restriction_\xi]$ , otherwise  $x \in M[G \restriction_\xi]$ .

Choose  $n < \omega, p_{s \smallfrown 0}, p_{s \smallfrown 1} \in \text{Col}(\omega, \eta)$  such that  $p_{s \smallfrown 0} \leq p_s, p_{s \smallfrown 1} \leq p_s, p_{s \smallfrown 0} \Vdash \tau(\check{n}) = \check{m}, p_{s \smallfrown 1} \Vdash \tau(\check{n}) = \check{k}$  where  $m \neq k$ . Then  $p_{s \smallfrown 0}, p_{s \smallfrown 1} \in D_{\text{dom}(s)}$ , for every  $z \in {}^\omega 2, \{q : \exists l, q \geq p_{z \restriction l}\}$  is  $\text{Col}(\omega, \eta)$ -generic over  $M[G \restriction_\xi]$ , call it  $G_z^*$ .

Then  $P := \{\tau^{G_z^*} : z \in {}^\omega 2 \cap M[G]\}$  is a perfect set in  $M[G]$ , because: Clearly it is not empty. We pick  $z' \in {}^\omega 2 \cap M[G]$ , if  $z \restriction_n = z' \restriction_n$  then  $\tau^{G_{z \restriction_n}^*} = \tau^{G_{z' \restriction_n}^*}$  thus every point is an accumulation point. Let  $\tau^{G_{z_n}^*}$

convergence to  $y$ , which means  $\forall k, \exists n(k), \forall n \geq n(k), z_n \upharpoonright_k = z_{n(k)} \upharpoonright_k$ . Let  $z = \bigcup_{k < \omega} z_{n(k)} \upharpoonright_k$  then  $y = \tau^{G_z^*}$  this gives closedness.

By the choice of  $\tau, P \subset A$ . This gives the perfect set property. ■